Classification with Neural Networks - Case Study

Let's consider an example of classification problem solved by using Neural Networks.

Dataset Description

Breast cancer Wisconsin (diagnostic) dataset

Data Set Characteristics:

```
    :Number of Instances: 569

    :Number of Attributes: 30 numeric, predictive attributes and
    the class

    :Attribute Information:
        - radius (mean of distances from center to points on the
    perimeter)
        - texture (standard deviation of gray-scale values)
        - perimeter
        - area
        - smoothness (local variation in radius lengths)
        - compactness (perimeter^2 / area - 1.0)
        - concavity (severity of concave portions of the contour)
        - concave points (number of concave portions of the
    contour)
        - symmetry
        - fractal dimension ("coastline approximation" - 1)

        The mean, standard error, and "worst" or largest (mean of
    the three
        largest values) of these features were computed for each
    image,
        resulting in 30 features.  For instance, field 3 is Mean
    Radius, field
        13 is Radius SE, field 23 is Worst Radius.

        - class:
                - WDBC-Malignant
                - WDBC-Benign

    :Summary Statistics:
```

```
================================== ====== ======
                                    Min    Max
================================== ====== ======
radius (mean):                     6.981  28.11
texture (mean):                    9.71   39.28
perimeter (mean):                  43.79  188.5
area (mean):                       143.5  2501.0
smoothness (mean):                 0.053  0.163
compactness (mean):                0.019  0.345
concavity (mean):                  0.0    0.427
concave points (mean):             0.0    0.201
symmetry (mean):                   0.106  0.304
fractal dimension (mean):          0.05   0.097
radius (standard error):           0.112  2.873
texture (standard error):          0.36   4.885
perimeter (standard error):        0.757  21.98
area (standard error):             6.802  542.2
smoothness (standard error):       0.002  0.031
compactness (standard error):      0.002  0.135
concavity (standard error):        0.0    0.396
concave points (standard error):   0.0    0.053
symmetry (standard error):         0.008  0.079
fractal dimension (standard error): 0.001 0.03
radius (worst):                    7.93   36.04
texture (worst):                   12.02  49.54
perimeter (worst):                 50.41  251.2
area (worst):                      185.2  4254.0
smoothness (worst):                0.071  0.223
compactness (worst):               0.027  1.058
concavity (worst):                 0.0    1.252
concave points (worst):            0.0    0.291
symmetry (worst):                  0.156  0.664
fractal dimension (worst):         0.055  0.208
================================== ====== ======

:Missing Attribute Values: None

:Class Distribution: 212 - Malignant, 357 - Benign

:Creator:  Dr. William H. Wolberg, W. Nick Street, Olvi L.
Mangasarian

:Donor: Nick Street

:Date: November, 1995
```

This is a copy of UCI ML Breast Cancer Wisconsin (Diagnostic) datasets. https://goo.gl/U2Uwz2

Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. They describe characteristics of the cell nuclei present in the

image.

Separating plane described above was obtained using Multisurface Method-Tree (MSM-T) [K. P. Bennett, "Decision Tree Construction Via Linear Programming." Proceedings of the 4th Midwest Artificial Intelligence and Cognitive Science Society, pp. 97-101, 1992], a classification method which uses linear programming to construct a decision tree. Relevant features were selected using an exhaustive search in the space of 1-4 features and 1-3 separating planes.

The actual linear program used to obtain the separating plane in the 3-dimensional space is that described in: [K. P. Bennett and O. L. Mangasarian: "Robust Linear Programming Discrimination of Two Linearly Inseparable Sets", Optimization Methods and Software 1, 1992, 23-34].

This database is also available through the UW CS ftp server:

ftp ftp.cs.wisc.edu cd math-prog/cpo-dataset/machine-learn/WDBC/

.. topic:: References

- W.N. Street, W.H. Wolberg and O.L. Mangasarian. Nuclear feature extraction for breast tumor diagnosis. IS&T/SPIE 1993 International Symposium on Electronic Imaging: Science and Technology, volume 1905, pages 861-870, San Jose, CA, 1993.
- O.L. Mangasarian, W.N. Street and W.H. Wolberg. Breast cancer diagnosis and prognosis via linear programming. Operations Research, 43(4), pages 570-577, July-August 1995.
- W.H. Wolberg, W.N. Street, and O.L. Mangasarian. Machine learning techniques to diagnose breast cancer from fine-needle aspirates. Cancer Letters 77 (1994) 163-171.

Import Data Analysis and Visualization Libraries

In [1]:
```
%pip install numpy pandas matplotlib seaborn openpyxl
%pip install ipywidgets

import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline
```

```
Requirement already satisfied: numpy in /usr/local/python/3.14.2/lib/python
3.14/site-packages (2.5.4)
Requirement already satisfied: pandas in /usr/local/python/3.14.2/lib/python
3.14/site-packages (3.1.0)
Requirement already satisfied: matplotlib in /usr/local/python/3.14.2/lib/py
thon3.14/site-packages (3.11.2)
Requirement already satisfied: seaborn in /usr/local/python/3.14.2/lib/pytho
n3.14/site-packages (0.13.2)
Requirement already satisfied: openpyxl in /usr/local/python/3.14.2/lib/pyth
on3.14/site-packages (3.1.5)
Requirement already satisfied: python-dateutil>=2.9.0 in /home/codespace/.lo
cal/lib/python3.14/site-packages (from pandas) (3.0.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/python/3.14.2/
lib/python3.14/site-packages (from matplotlib) (1.4.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/python/3.14.2/lib/
python3.14/site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in /usr/local/python/3.14.
2/lib/python3.14/site-packages (from matplotlib) (4.68.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/python/3.14.
2/lib/python3.14/site-packages (from matplotlib) (1.5.1)
Requirement already satisfied: packaging>=20.0 in /home/codespace/.local/li
b/python3.14/site-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=9 in /usr/local/python/3.14.2/lib/pyt
hon3.14/site-packages (from matplotlib) (13.0.0)
Requirement already satisfied: pyparsing>=3 in /usr/local/python/3.14.2/lib/
python3.14/site-packages (from matplotlib) (3.3.3)
Requirement already satisfied: et-xmlfile in /usr/local/python/3.14.2/lib/py
thon3.14/site-packages (from openpyxl) (2.0.0)
Requirement already satisfied: six>=1.6 in /home/codespace/.local/lib/python
3.14/site-packages (from python-dateutil>=2.9.0->pandas) (1.17.0)
Requirement already satisfied: tzdata in /home/codespace/.local/lib/python3.
14/site-packages (from python-dateutil>=2.9.0->pandas) (2026.6)
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: ipywidgets in /usr/local/python/3.14.2/lib/py
thon3.14/site-packages (8.1.9)
Requirement already satisfied: comm>=0.1.3 in /home/codespace/.local/lib/pyt
hon3.14/site-packages (from ipywidgets) (0.2.3)
Requirement already satisfied: ipython>=6.1.0 in /home/codespace/.local/lib/
python3.14/site-packages (from ipywidgets) (9.17.1)
Requirement already satisfied: traitlets>=4.3.1 in /home/codespace/.local/li
b/python3.14/site-packages (from ipywidgets) (5.16.1)
Requirement already satisfied: widgetsnbextension~=4.0.16 in /usr/local/pyth
on/3.14.2/lib/python3.14/site-packages (from ipywidgets) (4.0.16)
Requirement already satisfied: jupyterlab_widgets~=3.0.17 in /usr/local/pyth
on/3.14.2/lib/python3.14/site-packages (from ipywidgets) (3.0.17)
Requirement already satisfied: ipython-pygments-lexers>=1.0.0 in /home/codes
pace/.local/lib/python3.14/site-packages (from ipython>=6.1.0->ipywidgets)
(1.1.1)
Requirement already satisfied: jedi>=0.18.2 in /home/codespace/.local/lib/py
thon3.14/site-packages (from ipython>=6.1.0->ipywidgets) (0.20.1)
Requirement already satisfied: matplotlib-inline>=0.1.6 in /home/codespace/.
local/lib/python3.14/site-packages (from ipython>=6.1.0->ipywidgets) (0.2.2)
Requirement already satisfied: pexpect>4.6 in /home/codespace/.local/lib/pyt
hon3.14/site-packages (from ipython>=6.1.0->ipywidgets) (4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in /home/codesp
ace/.local/lib/python3.14/site-packages (from ipython>=6.1.0->ipywidgets)
```

```
(3.0.53)
Requirement already satisfied: psutil>=7 in /home/codespace/.local/lib/pytho
n3.14/site-packages (from ipython>=6.1.0->ipywidgets) (7.2.2)
Requirement already satisfied: pygments>=2.14.0 in /home/codespace/.local/li
b/python3.14/site-packages (from ipython>=6.1.0->ipywidgets) (2.21.0)
Requirement already satisfied: stack_data>=0.6.0 in /home/codespace/.local/l
ib/python3.14/site-packages (from ipython>=6.1.0->ipywidgets) (0.6.3)
Requirement already satisfied: wcwidth>=0.1.4 in /home/codespace/.local/lib/
python3.14/site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython>=6.1.0
->ipywidgets) (0.9.4)
Requirement already satisfied: parso<0.9.0,>=0.8.7 in /home/codespace/.loca
l/lib/python3.14/site-packages (from jedi>=0.18.2->ipython>=6.1.0->ipywidget
s) (0.8.7)
Requirement already satisfied: ptyprocess>=0.5 in /home/codespace/.local/li
b/python3.14/site-packages (from pexpect>4.6->ipython>=6.1.0->ipywidgets)
(0.7.0)
Requirement already satisfied: executing>=1.2.0 in /home/codespace/.local/li
b/python3.14/site-packages (from stack_data>=0.6.0->ipython>=6.1.0->ipywidge
ts) (2.3.0)
Requirement already satisfied: asttokens>=2.1.0 in /home/codespace/.local/li
b/python3.14/site-packages (from stack_data>=0.6.0->ipython>=6.1.0->ipywidge
ts) (3.0.2)
Requirement already satisfied: pure-eval in /home/codespace/.local/lib/pytho
n3.14/site-packages (from stack_data>=0.6.0->ipython>=6.1.0->ipywidgets) (0.
2.4)
Note: you may need to restart the kernel to use updated packages.
```

Import Dataset

In [2]:
```
df = pd.read_csv('cancer_classification.csv')          # Import Dataset from
df.rename(columns={"benign_0__mal_1" : "benign_1__mal_0"}, inplace = True)
df.head()                                              # Show header
```

```
/tmp/ipykernel_39069/3614546604.py:2: Pandas4Warning: The inplace keyword in
DataFrame.rename is deprecated and will be removed in a future version (PDEP
-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inpl
ace for more details.
  df.rename(columns={"benign_0__mal_1" : "benign_1__mal_0"}, inplace = True)
# Rename column
```

Out[2]:

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	c
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	

5 rows × 31 columns

In [3]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 31 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   mean radius              569 non-null    float64
 1   mean texture             569 non-null    float64
 2   mean perimeter           569 non-null    float64
 3   mean area                569 non-null    float64
 4   mean smoothness          569 non-null    float64
 5   mean compactness         569 non-null    float64
 6   mean concavity           569 non-null    float64
 7   mean concave points      569 non-null    float64
 8   mean symmetry            569 non-null    float64
 9   mean fractal dimension   569 non-null    float64
 10  radius error             569 non-null    float64
 11  texture error            569 non-null    float64
 12  perimeter error          569 non-null    float64
 13  area error               569 non-null    float64
 14  smoothness error         569 non-null    float64
 15  compactness error        569 non-null    float64
 16  concavity error          569 non-null    float64
 17  concave points error     569 non-null    float64
 18  symmetry error           569 non-null    float64
 19  fractal dimension error  569 non-null    float64
 20  worst radius             569 non-null    float64
 21  worst texture            569 non-null    float64
 22  worst perimeter          569 non-null    float64
 23  worst area               569 non-null    float64
 24  worst smoothness         569 non-null    float64
 25  worst compactness        569 non-null    float64
 26  worst concavity          569 non-null    float64
 27  worst concave points     569 non-null    float64
 28  worst symmetry           569 non-null    float64
 29  worst fractal dimension  569 non-null    float64
 30  benign_1__mal_0          569 non-null    int64  
dtypes: float64(30), int64(1)
memory usage: 137.9 KB
```

In [4]:
```
df.describe().transpose()
```

Out[4]:

	count	mean	std	min	25%	
mean radius	569.0	14.127292	3.524049	6.981000	11.700000	13.3
mean texture	569.0	19.289649	4.301036	9.710000	16.170000	18.8
mean perimeter	569.0	91.969033	24.298981	43.790000	75.170000	86.2
mean area	569.0	654.889104	351.914129	143.500000	420.300000	551.1
mean smoothness	569.0	0.096360	0.014064	0.052630	0.086370	0.0
mean compactness	569.0	0.104341	0.052813	0.019380	0.064920	0.0
mean concavity	569.0	0.088799	0.079720	0.000000	0.029560	0.0
mean concave points	569.0	0.048919	0.038803	0.000000	0.020310	0.0
mean symmetry	569.0	0.181162	0.027414	0.106000	0.161900	0.1
mean fractal dimension	569.0	0.062798	0.007060	0.049960	0.057700	0.0
radius error	569.0	0.405172	0.277313	0.111500	0.232400	0.3
texture error	569.0	1.216853	0.551648	0.360200	0.833900	1.1
perimeter error	569.0	2.866059	2.021855	0.757000	1.606000	2.2
area error	569.0	40.337079	45.491006	6.802000	17.850000	24.5
smoothness error	569.0	0.007041	0.003003	0.001713	0.005169	0.0
compactness error	569.0	0.025478	0.017908	0.002252	0.013080	0.0
concavity error	569.0	0.031894	0.030186	0.000000	0.015090	0.0
concave points error	569.0	0.011796	0.006170	0.000000	0.007638	0.0
symmetry error	569.0	0.020542	0.008266	0.007882	0.015160	0.0
fractal dimension error	569.0	0.003795	0.002646	0.000895	0.002248	0.0
worst radius	569.0	16.269190	4.833242	7.930000	13.010000	14.9
worst texture	569.0	25.677223	6.146258	12.020000	21.080000	25.4
worst perimeter	569.0	107.261213	33.602542	50.410000	84.110000	97.6
worst area	569.0	880.583128	569.356993	185.200000	515.300000	686.5
worst smoothness	569.0	0.132369	0.022832	0.071170	0.116600	0.1
worst compactness	569.0	0.254265	0.157336	0.027290	0.147200	0.2
worst concavity	569.0	0.272188	0.208624	0.000000	0.114500	0.2

	count	mean	std	min	25%	
worst concave points	569.0	0.114606	0.065732	0.000000	0.064930	0.0
worst symmetry	569.0	0.290076	0.061867	0.156500	0.250400	0.2
worst fractal dimension	569.0	0.083946	0.018061	0.055040	0.071460	0.0
benign_1__mal_0	569.0	0.627417	0.483918	0.000000	0.000000	1.0

Exploratory Data Analysis

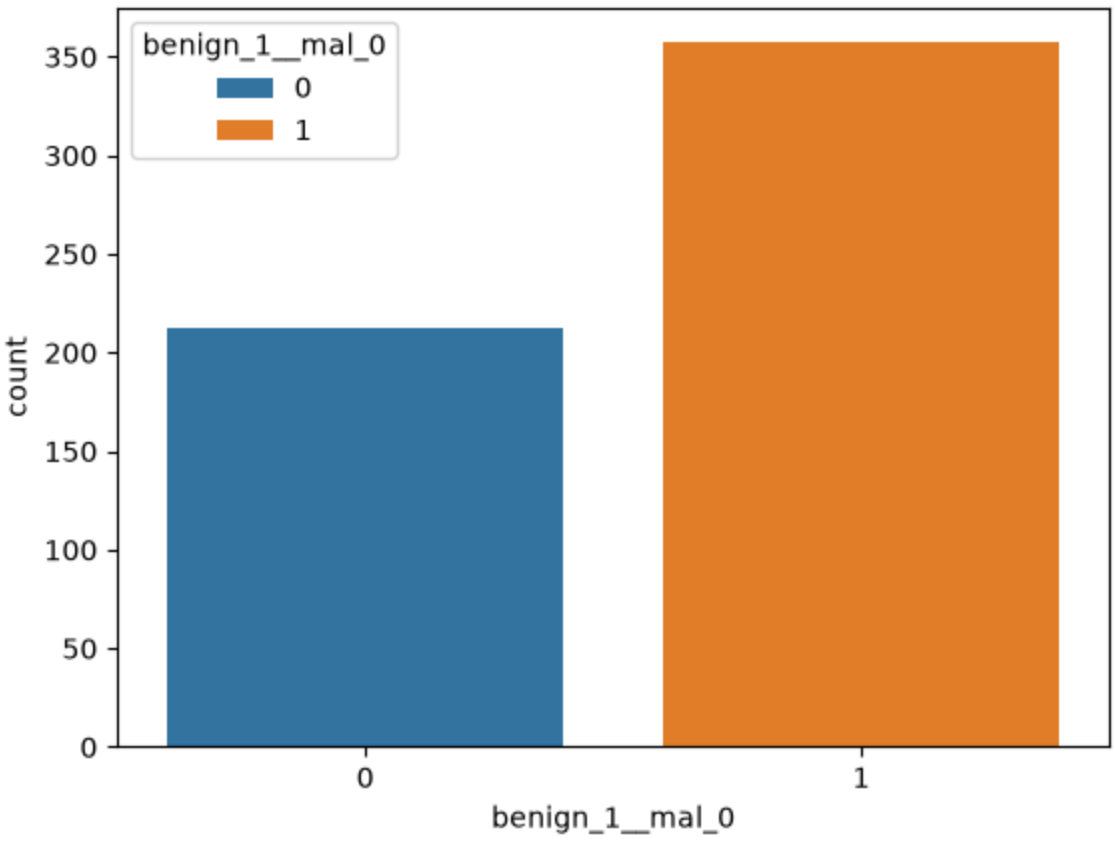
In [5]:
```
# Plot number of benign vs malignant
sns.countplot(x='benign_1__mal_0',data=df,hue='benign_1__mal_0')

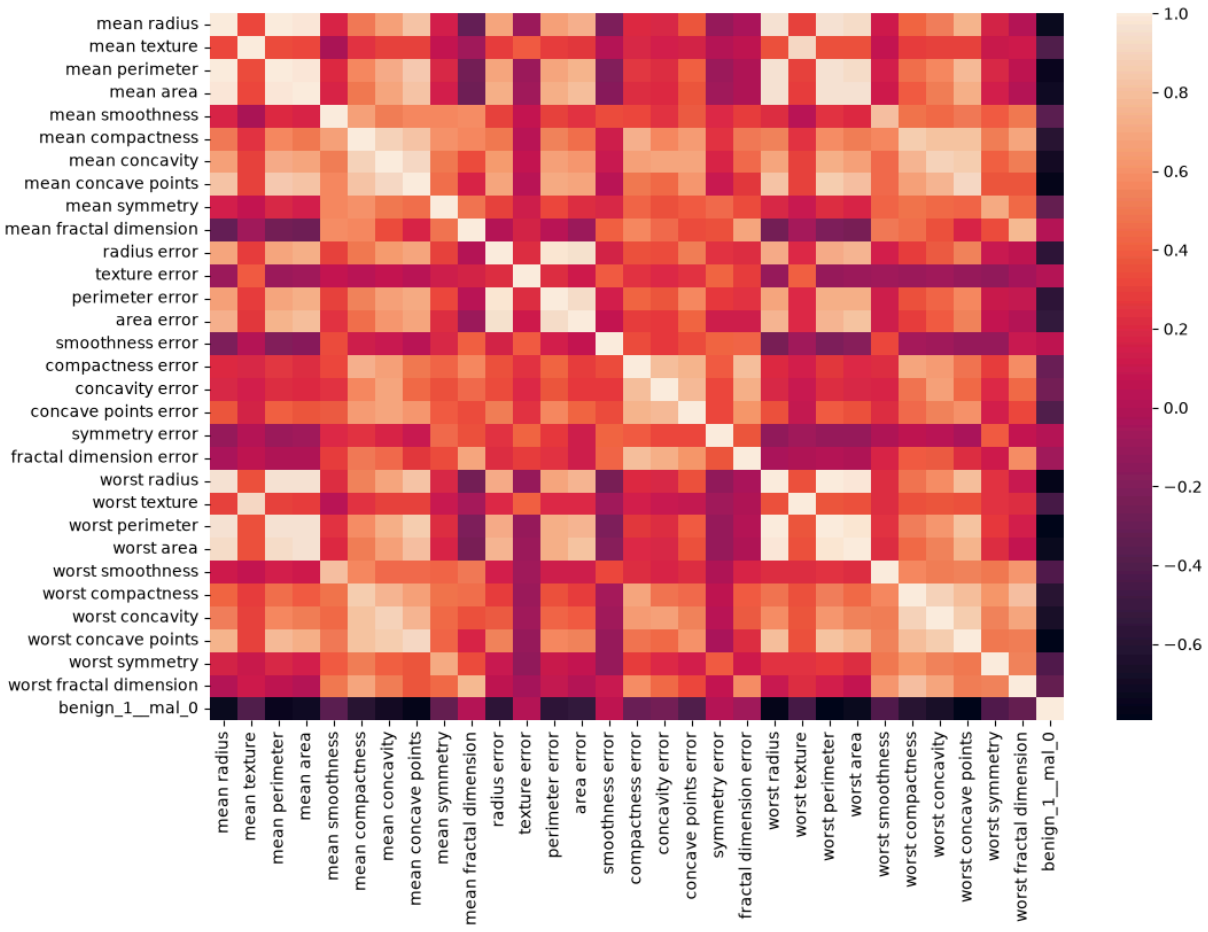
# Plot correlation heatmap
plt.figure(figsize=(12,8))
sns.heatmap(df.corr())

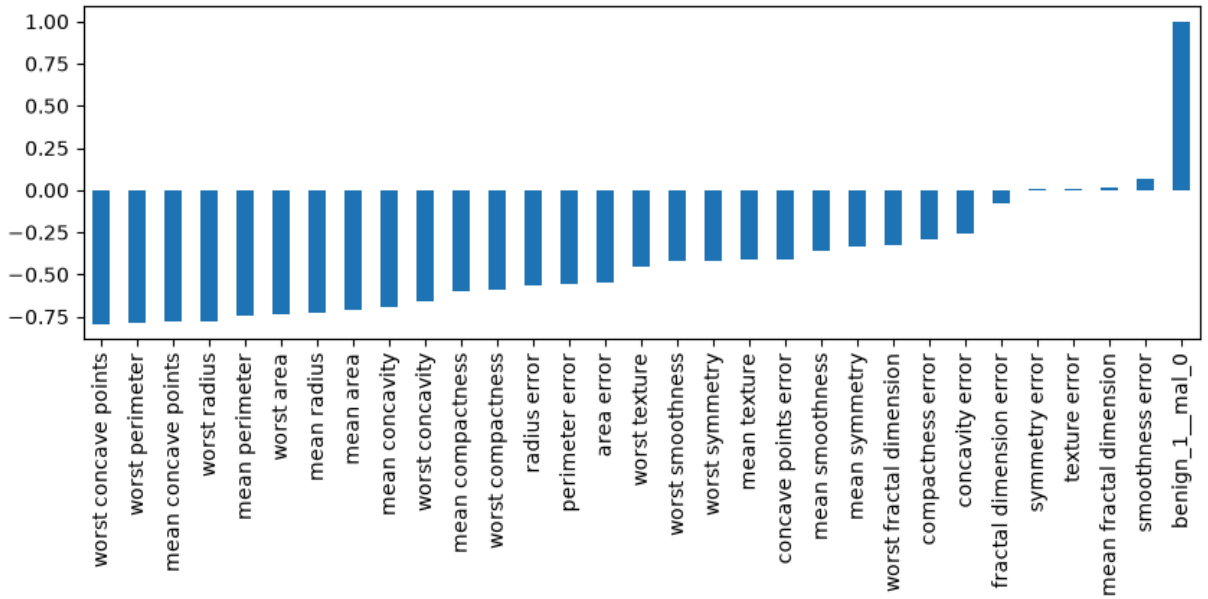
# Reorder correlation values for easier inspection
df.corr()['benign_1__mal_0'].sort_values(ascending = False)

# Plot these values
plt.figure(figsize=(10,3))                                    # Create a
df.corr()['benign_1__mal_0'].sort_values().plot(kind='bar')   # Barplot
```

Out[5]: <Axes: >







Train/Test Set Split

In [6]:
```
from sklearn.model_selection import train_test_split            # Import tra

# Features
X = df.drop('benign_1__mal_0',axis=1).values
```

```
# Target Variable
y = df['benign_1__mal_0'].values

# Perform split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, ra
print(X_train.shape, X_test.shape, y_train.shape, y_test.shape)
```

```
(426, 30) (143, 30) (426,) (143,)
```

Data Pre-Processing

Use StandardScaler() to perform input centering and input normalization.

Notice that it is sufficient to standardize just input features, while it is not necessary to standardize the target variable.

In [7]:
```
from sklearn.preprocessing import StandardScaler

# Train scaler
scaler = StandardScaler()
scaler.fit(X_train)

# Apply scaling
X_train = scaler.transform(X_train)
X_test = scaler.transform(X_test)
```

Neural Network Definition - TensorFlow & Keras

Creating NN Model

In [8]:
```
import tensorflow as tf

from tensorflow.keras.models import Sequential                          # Imp
from tensorflow.keras.layers import Dense                               # Imp
```

```
WARNING: All log messages before absl::InitializeLog() is called are written
to STDERR
I0000 00:00:1793454233.753442   39069 port.cc:153] oneDNN custom operations
are on. You may see slightly different numerical results due to floating-poi
nt round-off errors from different computation orders. To turn them off, set
the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1793454233.800236   39069 cudart_stub.cc:31] Could not find cuda
drivers on your machine, GPU will not be used.
I0000 00:00:1793454235.310074   39069 cpu_feature_guard.cc:227] This TensorF
low binary is optimized to use available CPU instructions in performance-cri
tical operations.
To enable the following instructions: AVX2 AVX512F AVX512_VNNI FMA, in other
operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING: All log messages before absl::InitializeLog() is called are written
to STDERR
I0000 00:00:1793454239.209874   39069 port.cc:153] oneDNN custom operations
are on. You may see slightly different numerical results due to floating-poi
nt round-off errors from different computation orders. To turn them off, set
the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1793454239.213252   39069 cudart_stub.cc:31] Could not find cuda
drivers on your machine, GPU will not be used.
```

In [9]:
```
# Generate model
model = Sequential()
model.add(Dense(units=30,activation='relu'))
model.add(Dense(units=15,activation='relu'))

# Final output node for prediction
model.add(Dense(units=1, activation='sigmoid'))                    # Ir

# For a binary classification problem
model.compile(loss='binary_crossentropy', optimizer='adam')        # Op
```

```
E0000 00:00:1793454240.266036   39069 cuda_platform.cc:57] failed call to cu
Init: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)
=== Source Location Trace: ===
external/xla/xla/stream_executor/cuda/cuda_status.cc:50
```

Choosing suitable Loss functions and Evaluation Metrics

Keep in mind what kind of problem you are trying to solve:

```
# For a multi-class classification problem
model.compile(optimizer='rmsprop',
              loss='categorical_crossentropy',
              metrics='accuracy')

# For a binary classification problem
model.compile(optimizer='rmsprop',
              loss='binary_crossentropy',
              metrics='accuracy')

# For a mean squared error regression problem
```

```
model.compile(optimizer='rmsprop',
              loss='mse')
```

Training the Model

Example One

In [ ]:
```
model.fit(x=X_train, y=y_train, epochs=1000, validation_data=(X_test, y_test
# overfitting because errors are different
```

```
Epoch 1/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - loss: 0.6887 - val_loss: 0.5331
Epoch 2/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.4493 - val_loss: 0.3716
Epoch 3/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.3048 - val_loss: 0.2679
Epoch 4/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.2179 - val_loss: 0.2029
Epoch 5/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.1677 - val_loss: 0.1639
Epoch 6/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.1384 - val_loss: 0.1398
Epoch 7/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1196 - val_loss: 0.1240
Epoch 8/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1066 - val_loss: 0.1131
Epoch 9/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0971 - val_loss: 0.1048
Epoch 10/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0896 - val_loss: 0.0985
Epoch 11/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0835 - val_loss: 0.0934
Epoch 12/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0784 - val_loss: 0.0891
Epoch 13/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0742 - val_loss: 0.0855
Epoch 14/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0705 - val_loss: 0.0823
Epoch 15/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0673 - val_loss: 0.0794
Epoch 16/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0645 - val_loss: 0.0769
Epoch 17/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0619 - val_loss: 0.0747
Epoch 18/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0595 - val_loss: 0.0727
Epoch 19/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0573 - val_loss: 0.0709
Epoch 20/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0552 - val_loss: 0.0690
Epoch 21/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0533 - val_loss: 0.0673
Epoch 22/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0514 - val_loss: 0.0657
Epoch 23/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0497 - val_loss: 0.0643
Epoch 24/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0481 - val_loss: 0.0628
Epoch 25/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0465 - val_loss: 0.0615
Epoch 26/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0450 - val_loss: 0.0604
Epoch 27/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0436 - val_loss: 0.0592
Epoch 28/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0422 - val_loss: 0.0583
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Epoch 29/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0408 - val_loss: 0.0573
Epoch 30/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0394 - val_loss: 0.0563
Epoch 31/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0382 - val_loss: 0.0556
Epoch 32/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0369 - val_loss: 0.0546
Epoch 33/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0357 - val_loss: 0.0537
Epoch 34/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0345 - val_loss: 0.0528
Epoch 35/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0334 - val_loss: 0.0519
Epoch 36/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0322 - val_loss: 0.0511
Epoch 37/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0312 - val_loss: 0.0502
Epoch 38/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0301 - val_loss: 0.0496
Epoch 39/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0290 - val_loss: 0.0488
Epoch 40/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0279 - val_loss: 0.0480
Epoch 41/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0269 - val_loss: 0.0475
Epoch 42/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0259 - val_loss: 0.0470
Epoch 43/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0250 - val_loss: 0.0463
Epoch 44/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0241 - val_loss: 0.0456
Epoch 45/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0232 - val_loss: 0.0448
Epoch 46/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0224 - val_loss: 0.0443
Epoch 47/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0215 - val_loss: 0.0437
Epoch 48/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0207 - val_loss: 0.0431
Epoch 49/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0200 - val_loss: 0.0425
Epoch 50/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0192 - val_loss: 0.0420
Epoch 51/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0186 - val_loss: 0.0413
Epoch 52/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0179 - val_loss: 0.0407
Epoch 53/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0173 - val_loss: 0.0402
Epoch 54/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0166 - val_loss: 0.0395
Epoch 55/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 0.0161 - val_loss: 0.0391
Epoch 56/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0155 - val_loss: 0.0387
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Epoch 57/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0150 - val_loss: 0.0382
Epoch 58/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0145 - val_loss: 0.0378
Epoch 59/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0140 - val_loss: 0.0373
Epoch 60/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0135 - val_loss: 0.0369
Epoch 61/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0131 - val_loss: 0.0367
Epoch 62/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0126 - val_loss: 0.0364
Epoch 63/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0122 - val_loss: 0.0360
Epoch 64/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0118 - val_loss: 0.0355
Epoch 65/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - loss: 0.0114 - val_loss: 0.0353
Epoch 66/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0111 - val_loss: 0.0349
Epoch 67/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0107 - val_loss: 0.0348
Epoch 68/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0104 - val_loss: 0.0344
Epoch 69/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0101 - val_loss: 0.0341
Epoch 70/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0097 - val_loss: 0.0338
Epoch 71/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0094 - val_loss: 0.0336
Epoch 72/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0091 - val_loss: 0.0333
Epoch 73/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0089 - val_loss: 0.0331
Epoch 74/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0086 - val_loss: 0.0328
Epoch 75/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0083 - val_loss: 0.0327
Epoch 76/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0081 - val_loss: 0.0325
Epoch 77/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0078 - val_loss: 0.0323
Epoch 78/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0076 - val_loss: 0.0322
Epoch 79/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0074 - val_loss: 0.0320
Epoch 80/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0071 - val_loss: 0.0317
Epoch 81/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0069 - val_loss: 0.0317
Epoch 82/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0067 - val_loss: 0.0315
Epoch 83/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0065 - val_loss: 0.0314
Epoch 84/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0063 - val_loss: 0.0312
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Epoch 85/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0061 - val_loss: 0.0311
Epoch 86/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0059 - val_loss: 0.0310
Epoch 87/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0057 - val_loss: 0.0308
Epoch 88/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0056 - val_loss: 0.0307
Epoch 89/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0054 - val_loss: 0.0305
Epoch 90/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0052 - val_loss: 0.0305
Epoch 91/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0051 - val_loss: 0.0302
Epoch 92/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0049 - val_loss: 0.0301
Epoch 93/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0048 - val_loss: 0.0302
Epoch 94/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0046 - val_loss: 0.0301
Epoch 95/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0045 - val_loss: 0.0302
Epoch 96/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0043 - val_loss: 0.0300
Epoch 97/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0042 - val_loss: 0.0300
Epoch 98/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0040 - val_loss: 0.0301
Epoch 99/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0039 - val_loss: 0.0302
Epoch 100/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0038 - val_loss: 0.0302
Epoch 101/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0036 - val_loss: 0.0302
Epoch 102/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0035 - val_loss: 0.0303
Epoch 103/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0034 - val_loss: 0.0302
Epoch 104/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0033 - val_loss: 0.0302
Epoch 105/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0032 - val_loss: 0.0301
Epoch 106/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0031 - val_loss: 0.0302
Epoch 107/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0030 - val_loss: 0.0302
Epoch 108/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0029 - val_loss: 0.0303
Epoch 109/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0028 - val_loss: 0.0303
Epoch 110/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0028 - val_loss: 0.0303
Epoch 111/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0027 - val_loss: 0.0305
Epoch 112/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0026 - val_loss: 0.0304
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Epoch 113/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0025 - val_loss: 0.0305
Epoch 114/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0025 - val_loss: 0.0305
Epoch 115/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0024 - val_loss: 0.0307
Epoch 116/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0023 - val_loss: 0.0308
Epoch 117/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0023 - val_loss: 0.0309
Epoch 118/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0022 - val_loss: 0.0309
Epoch 119/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0022 - val_loss: 0.0310
Epoch 120/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0021 - val_loss: 0.0311
Epoch 121/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0021 - val_loss: 0.0312
Epoch 122/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0020 - val_loss: 0.0313
Epoch 123/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0020 - val_loss: 0.0313
Epoch 124/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0019 - val_loss: 0.0314
Epoch 125/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0019 - val_loss: 0.0316
Epoch 126/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0018 - val_loss: 0.0317
Epoch 127/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0018 - val_loss: 0.0317
Epoch 128/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0017 - val_loss: 0.0318
Epoch 129/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0017 - val_loss: 0.0319
Epoch 130/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0017 - val_loss: 0.0319
Epoch 131/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0016 - val_loss: 0.0320
Epoch 132/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0016 - val_loss: 0.0321
Epoch 133/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0015 - val_loss: 0.0322
Epoch 134/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0015 - val_loss: 0.0323
Epoch 135/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0015 - val_loss: 0.0324
Epoch 136/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0014 - val_loss: 0.0325
Epoch 137/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0014 - val_loss: 0.0326
Epoch 138/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0014 - val_loss: 0.0327
Epoch 139/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0013 - val_loss: 0.0327
Epoch 140/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0013 - val_loss: 0.0329
```

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Epoch 141/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0013 - val_loss: 0.0330
Epoch 142/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0012 - val_loss: 0.0330
Epoch 143/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0012 - val_loss: 0.0331
Epoch 144/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0012 - val_loss: 0.0332
Epoch 145/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0012 - val_loss: 0.0333
Epoch 146/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0011 - val_loss: 0.0335
Epoch 147/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0011 - val_loss: 0.0335
Epoch 148/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0011 - val_loss: 0.0336
Epoch 149/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0011 - val_loss: 0.0337
Epoch 150/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0011 - val_loss: 0.0338
Epoch 151/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0010 - val_loss: 0.0339
Epoch 152/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0010 - val_loss: 0.0340
Epoch 153/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 9.8883e-04 - val_loss: 0.0341
Epoch 154/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.7166e-04 - val_loss: 0.0342
Epoch 155/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 9.5899e-04 - val_loss: 0.0342
Epoch 156/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 9.3094e-04 - val_loss: 0.0344
Epoch 157/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.1268e-04 - val_loss: 0.0345
Epoch 158/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 8.9442e-04 - val_loss: 0.0346
Epoch 159/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 8.7396e-04 - val_loss: 0.0347
Epoch 160/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 8.5722e-04 - val_loss: 0.0349
Epoch 161/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 8.3947e-04 - val_loss: 0.0349
Epoch 162/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 8.1801e-04 - val_loss: 0.0351
Epoch 163/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.0333e-04 - val_loss: 0.0351
Epoch 164/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.8744e-04 - val_loss: 0.0353
Epoch 165/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.7041e-04 - val_loss: 0.0354
Epoch 166/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.5241e-04 - val_loss: 0.0355
Epoch 167/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.3808e-04 - val_loss: 0.0355
Epoch 168/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.2284e-04 - val_loss: 0.0357
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Epoch 169/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.0891e-04 - val_loss: 0.0357
Epoch 170/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.9231e-04 - val_loss: 0.0358
Epoch 171/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.8272e-04 - val_loss: 0.0359
Epoch 172/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.6730e-04 - val_loss: 0.0359
Epoch 173/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 6.5292e-04 - val_loss: 0.036
0
Epoch 174/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.4035e-04 - val_loss: 0.0360
Epoch 175/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.2745e-04 - val_loss: 0.0361
Epoch 176/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.1428e-04 - val_loss: 0.0362
Epoch 177/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.0321e-04 - val_loss: 0.0363
Epoch 178/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.9251e-04 - val_loss: 0.0363
Epoch 179/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.7934e-04 - val_loss: 0.0365
Epoch 180/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 5.7002e-04 - val_loss: 0.0365
Epoch 181/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.5860e-04 - val_loss: 0.0366
Epoch 182/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.4712e-04 - val_loss: 0.0367
Epoch 183/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 5.3899e-04 - val_loss: 0.0366
Epoch 184/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.2698e-04 - val_loss: 0.0368
Epoch 185/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 5.1812e-04 - val_loss: 0.0368
Epoch 186/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.0761e-04 - val_loss: 0.0369
Epoch 187/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.9864e-04 - val_loss: 0.0369
Epoch 188/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.8953e-04 - val_loss: 0.0370
Epoch 189/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.8090e-04 - val_loss: 0.0371
Epoch 190/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.7239e-04 - val_loss: 0.0372
Epoch 191/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.6301e-04 - val_loss: 0.0371
Epoch 192/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.5642e-04 - val_loss: 0.0373
Epoch 193/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.4943e-04 - val_loss: 0.0374
Epoch 194/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.3993e-04 - val_loss: 0.0375
Epoch 195/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.3122e-04 - val_loss: 0.0376
Epoch 196/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 4.2546e-04 - val_loss: 0.0375
Epoch 197/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 4.1784e-04 - val_loss: 0.0376
Epoch 198/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 4.1082e-04 - val_loss: 0.0378
Epoch 199/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.0344e-04 - val_loss: 0.0378
Epoch 200/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.9600e-04 - val_loss: 0.0379
Epoch 201/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 3.9074e-04 - val_loss: 0.0380
Epoch 202/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 3.8288e-04 - val_loss: 0.0380
Epoch 203/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 3.7725e-04 - val_loss: 0.0381
Epoch 204/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.7087e-04 - val_loss: 0.0381
Epoch 205/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.6533e-04 - val_loss: 0.0382
Epoch 206/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.5884e-04 - val_loss: 0.0383
Epoch 207/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.5274e-04 - val_loss: 0.0384
Epoch 208/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 3.4761e-04 - val_loss: 0.0384
Epoch 209/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.4103e-04 - val_loss: 0.0385
Epoch 210/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 3.3545e-04 - val_loss: 0.0386
Epoch 211/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - loss: 3.2999e-04 - val_loss: 0.038
6
Epoch 212/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 3.2503e-04 - val_loss: 0.0386
Epoch 213/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.2051e-04 - val_loss: 0.0388
Epoch 214/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.1473e-04 - val_loss: 0.0388
Epoch 215/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 3.0981e-04 - val_loss: 0.038
8
Epoch 216/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 3.0439e-04 - val_loss: 0.039
0
Epoch 217/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 3.0011e-04 - val_loss: 0.039
0
Epoch 218/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 2.9533e-04 - val_loss: 0.0391
Epoch 219/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 28ms/step - loss: 2.9048e-04 - val_loss: 0.039
1
Epoch 220/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 2.8656e-04 - val_loss: 0.039
2
Epoch 221/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.8225e-04 - val_loss: 0.0393
Epoch 222/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7694e-04 - val_loss: 0.0394
Epoch 223/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7345e-04 - val_loss: 0.0393
Epoch 224/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.6874e-04 - val_loss: 0.0395
Epoch 225/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.6539e-04 - val_loss: 0.0396
Epoch 226/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.6069e-04 - val_loss: 0.0396
Epoch 227/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 2.5741e-04 - val_loss: 0.0396
Epoch 228/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.5278e-04 - val_loss: 0.0398
Epoch 229/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.4968e-04 - val_loss: 0.0398
Epoch 230/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.4511e-04 - val_loss: 0.0399
Epoch 231/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.4335e-04 - val_loss: 0.0399
Epoch 232/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.3864e-04 - val_loss: 0.0400
Epoch 233/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.3466e-04 - val_loss: 0.0401
Epoch 234/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 2.3182e-04 - val_loss: 0.040
1
Epoch 235/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.2829e-04 - val_loss: 0.0402
Epoch 236/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.2494e-04 - val_loss: 0.0402
Epoch 237/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.2157e-04 - val_loss: 0.0404
Epoch 238/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.1819e-04 - val_loss: 0.0404
Epoch 239/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.1504e-04 - val_loss: 0.0405
Epoch 240/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 2.1223e-04 - val_loss: 0.0405
Epoch 241/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0934e-04 - val_loss: 0.0405
Epoch 242/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.0628e-04 - val_loss: 0.0407
Epoch 243/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - loss: 2.0362e-04 - val_loss: 0.040
8
Epoch 244/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0026e-04 - val_loss: 0.0408
Epoch 245/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9729e-04 - val_loss: 0.0408
Epoch 246/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 1.9461e-04 - val_loss: 0.0409
Epoch 247/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.9232e-04 - val_loss: 0.040
9
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Epoch 248/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 1.8906e-04 - val_loss: 0.041
0
Epoch 249/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 1.8724e-04 - val_loss: 0.0410
Epoch 250/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1.8454e-04 - val_loss: 0.0411
Epoch 251/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 1.8153e-04 - val_loss: 0.041
2
Epoch 252/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7990e-04 - val_loss: 0.0412
Epoch 253/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7658e-04 - val_loss: 0.0413
Epoch 254/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1.7417e-04 - val_loss: 0.0414
Epoch 255/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7189e-04 - val_loss: 0.0414
Epoch 256/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 1.6977e-04 - val_loss: 0.0415
Epoch 257/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1.6711e-04 - val_loss: 0.0416
Epoch 258/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.6533e-04 - val_loss: 0.0416
Epoch 259/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6306e-04 - val_loss: 0.0417
Epoch 260/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6074e-04 - val_loss: 0.0417
Epoch 261/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5858e-04 - val_loss: 0.0418
Epoch 262/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5621e-04 - val_loss: 0.0419
Epoch 263/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5463e-04 - val_loss: 0.0419
Epoch 264/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5233e-04 - val_loss: 0.0419
Epoch 265/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5025e-04 - val_loss: 0.0420
Epoch 266/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.4839e-04 - val_loss: 0.0420
Epoch 267/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 1.4650e-04 - val_loss: 0.0421
Epoch 268/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 1.4452e-04 - val_loss: 0.042
1
Epoch 269/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4293e-04 - val_loss: 0.0422
Epoch 270/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4069e-04 - val_loss: 0.0423
Epoch 271/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1.3883e-04 - val_loss: 0.0423
Epoch 272/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1.3749e-04 - val_loss: 0.0424
Epoch 273/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.3566e-04 - val_loss: 0.0425
Epoch 274/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 1.3347e-04 - val_loss: 0.042
5
Epoch 275/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3197e-04 - val_loss: 0.0425
Epoch 276/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3052e-04 - val_loss: 0.0426
Epoch 277/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2864e-04 - val_loss: 0.0427
Epoch 278/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2701e-04 - val_loss: 0.0428
Epoch 279/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.2542e-04 - val_loss: 0.0427
Epoch 280/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.2385e-04 - val_loss: 0.0429
Epoch 281/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.2227e-04 - val_loss: 0.0429
Epoch 282/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 1.2084e-04 - val_loss: 0.042
9
Epoch 283/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.1899e-04 - val_loss: 0.0430
Epoch 284/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1.1763e-04 - val_loss: 0.0431
Epoch 285/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1625e-04 - val_loss: 0.0431
Epoch 286/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1459e-04 - val_loss: 0.0432
Epoch 287/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1335e-04 - val_loss: 0.0433
Epoch 288/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 1.1204e-04 - val_loss: 0.043
2
Epoch 289/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1047e-04 - val_loss: 0.0434
Epoch 290/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0903e-04 - val_loss: 0.0434
Epoch 291/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0784e-04 - val_loss: 0.0435
Epoch 292/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0664e-04 - val_loss: 0.0435
Epoch 293/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0508e-04 - val_loss: 0.0436
Epoch 294/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 1.0374e-04 - val_loss: 0.043
6
Epoch 295/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0282e-04 - val_loss: 0.0437
Epoch 296/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0140e-04 - val_loss: 0.0438
Epoch 297/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0008e-04 - val_loss: 0.0438
Epoch 298/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.8808e-05 - val_loss: 0.0439
Epoch 299/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.7924e-05 - val_loss: 0.0438
Epoch 300/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 9.6473e-05 - val_loss: 0.0440
Epoch 301/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.5566e-05 - val_loss: 0.0440
Epoch 302/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.4415e-05 - val_loss: 0.0441
Epoch 303/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.3054e-05 - val_loss: 0.0441
Epoch 304/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.1995e-05 - val_loss: 0.0441
Epoch 305/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.0951e-05 - val_loss: 0.0442
Epoch 306/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 8.9681e-05 - val_loss: 0.0443
Epoch 307/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.8628e-05 - val_loss: 0.0444
Epoch 308/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.7830e-05 - val_loss: 0.0444
Epoch 309/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.6610e-05 - val_loss: 0.0444
Epoch 310/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.5790e-05 - val_loss: 0.0445
Epoch 311/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.4740e-05 - val_loss: 0.0445
Epoch 312/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.3580e-05 - val_loss: 0.0446
Epoch 313/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.2646e-05 - val_loss: 0.0446
Epoch 314/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.1792e-05 - val_loss: 0.0446
Epoch 315/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.0721e-05 - val_loss: 0.0448
Epoch 316/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 7.9573e-05 - val_loss: 0.0448
Epoch 317/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.8929e-05 - val_loss: 0.0450
Epoch 318/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 7.8264e-05 - val_loss: 0.0449
Epoch 319/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 7.6997e-05 - val_loss: 0.0450
Epoch 320/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.6267e-05 - val_loss: 0.0450
Epoch 321/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.5321e-05 - val_loss: 0.0451
Epoch 322/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.4437e-05 - val_loss: 0.0452
Epoch 323/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.3553e-05 - val_loss: 0.0452
Epoch 324/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.2702e-05 - val_loss: 0.0453
Epoch 325/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - loss: 7.1982e-05 - val_loss: 0.045
3
Epoch 326/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 7.1164e-05 - val_loss: 0.0453
Epoch 327/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.0236e-05 - val_loss: 0.0454
```

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Epoch 328/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 6.9376e-05 - val_loss: 0.0455
Epoch 329/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.8675e-05 - val_loss: 0.0456
Epoch 330/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.8045e-05 - val_loss: 0.0456
Epoch 331/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.7104e-05 - val_loss: 0.0456
Epoch 332/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.6408e-05 - val_loss: 0.0457
Epoch 333/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.5716e-05 - val_loss: 0.0457
Epoch 334/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 6.4893e-05 - val_loss: 0.045
9
Epoch 335/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.4312e-05 - val_loss: 0.0458
Epoch 336/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 6.3408e-05 - val_loss: 0.0459
Epoch 337/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 6.2672e-05 - val_loss: 0.0460
Epoch 338/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 6.2048e-05 - val_loss: 0.0460
Epoch 339/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.1372e-05 - val_loss: 0.0460
Epoch 340/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 6.0696e-05 - val_loss: 0.0462
Epoch 341/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 6.0021e-05 - val_loss: 0.046
2
Epoch 342/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.9325e-05 - val_loss: 0.0462
Epoch 343/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.8602e-05 - val_loss: 0.0463
Epoch 344/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.8124e-05 - val_loss: 0.0463
Epoch 345/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.7441e-05 - val_loss: 0.0464
Epoch 346/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 5.6850e-05 - val_loss: 0.0465
Epoch 347/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.6089e-05 - val_loss: 0.0465
Epoch 348/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.5657e-05 - val_loss: 0.0465
Epoch 349/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.4918e-05 - val_loss: 0.0466
Epoch 350/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.4304e-05 - val_loss: 0.0466
Epoch 351/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.3797e-05 - val_loss: 0.0466
Epoch 352/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.3170e-05 - val_loss: 0.0467
Epoch 353/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.2562e-05 - val_loss: 0.0468
Epoch 354/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.2077e-05 - val_loss: 0.0468
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Epoch 355/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.1440e-05 - val_loss: 0.0469
Epoch 356/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.0969e-05 - val_loss: 0.0469
Epoch 357/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.0367e-05 - val_loss: 0.0469
Epoch 358/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.9856e-05 - val_loss: 0.0471
Epoch 359/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.9347e-05 - val_loss: 0.0471
Epoch 360/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.8735e-05 - val_loss: 0.0471
Epoch 361/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.8204e-05 - val_loss: 0.0472
Epoch 362/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 4.7794e-05 - val_loss: 0.0472
Epoch 363/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.7295e-05 - val_loss: 0.0474
Epoch 364/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.6721e-05 - val_loss: 0.0474
Epoch 365/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.6255e-05 - val_loss: 0.0473
Epoch 366/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 4.5727e-05 - val_loss: 0.0473
Epoch 367/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.5208e-05 - val_loss: 0.0475
Epoch 368/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.4815e-05 - val_loss: 0.0475
Epoch 369/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.4360e-05 - val_loss: 0.0476
Epoch 370/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.3860e-05 - val_loss: 0.0477
Epoch 371/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.3378e-05 - val_loss: 0.0476
Epoch 372/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.2971e-05 - val_loss: 0.0477
Epoch 373/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.2534e-05 - val_loss: 0.0477
Epoch 374/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.2025e-05 - val_loss: 0.0478
Epoch 375/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.1615e-05 - val_loss: 0.0478
Epoch 376/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.1164e-05 - val_loss: 0.0478
Epoch 377/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.0760e-05 - val_loss: 0.0479
Epoch 378/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 4.0230e-05 - val_loss: 0.0480
Epoch 379/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.9885e-05 - val_loss: 0.0480
Epoch 380/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.9489e-05 - val_loss: 0.0481
Epoch 381/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 3.9142e-05 - val_loss: 0.0481
Epoch 382/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 3.8673e-05 - val_loss: 0.0481
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Epoch 383/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 3.8213e-05 - val_loss: 0.0483
Epoch 384/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 3.7890e-05 - val_loss: 0.048
3
Epoch 385/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.7447e-05 - val_loss: 0.0483
Epoch 386/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.7136e-05 - val_loss: 0.0484
Epoch 387/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.6848e-05 - val_loss: 0.0483
Epoch 388/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.6247e-05 - val_loss: 0.0485
Epoch 389/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 3.5968e-05 - val_loss: 0.048
5
Epoch 390/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.5492e-05 - val_loss: 0.0485
Epoch 391/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.5282e-05 - val_loss: 0.0486
Epoch 392/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.4897e-05 - val_loss: 0.0487
Epoch 393/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.4469e-05 - val_loss: 0.0487
Epoch 394/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.4192e-05 - val_loss: 0.0487
Epoch 395/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.3777e-05 - val_loss: 0.0488
Epoch 396/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.3473e-05 - val_loss: 0.0487
Epoch 397/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.3143e-05 - val_loss: 0.0489
Epoch 398/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.2799e-05 - val_loss: 0.0489
Epoch 399/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.2591e-05 - val_loss: 0.0489
Epoch 400/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.2087e-05 - val_loss: 0.0490
Epoch 401/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.1761e-05 - val_loss: 0.0490
Epoch 402/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.1465e-05 - val_loss: 0.0491
Epoch 403/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.1222e-05 - val_loss: 0.0491
Epoch 404/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.0818e-05 - val_loss: 0.0493
Epoch 405/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.0546e-05 - val_loss: 0.0493
Epoch 406/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 3.0265e-05 - val_loss: 0.0493
Epoch 407/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 2.9861e-05 - val_loss: 0.0493
Epoch 408/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.9589e-05 - val_loss: 0.0494
Epoch 409/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.9460e-05 - val_loss: 0.0493
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Epoch 410/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 2.9013e-05 - val_loss: 0.049
4
Epoch 411/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.8670e-05 - val_loss: 0.0495
Epoch 412/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.8476e-05 - val_loss: 0.0495
Epoch 413/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.8219e-05 - val_loss: 0.0496
Epoch 414/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.7882e-05 - val_loss: 0.0496
Epoch 415/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7628e-05 - val_loss: 0.0496
Epoch 416/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7338e-05 - val_loss: 0.0498
Epoch 417/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7046e-05 - val_loss: 0.0498
Epoch 418/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.6935e-05 - val_loss: 0.0497
Epoch 419/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.6519e-05 - val_loss: 0.0499
Epoch 420/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 2.6246e-05 - val_loss: 0.049
9
Epoch 421/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 2.5943e-05 - val_loss: 0.050
0
Epoch 422/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 2.5781e-05 - val_loss: 0.050
0
Epoch 423/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.5547e-05 - val_loss: 0.0501
Epoch 424/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.5213e-05 - val_loss: 0.0501
Epoch 425/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 2.5017e-05 - val_loss: 0.0502
Epoch 426/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.4711e-05 - val_loss: 0.0502
Epoch 427/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.4533e-05 - val_loss: 0.0502
Epoch 428/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.4298e-05 - val_loss: 0.0503
Epoch 429/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.4027e-05 - val_loss: 0.0504
Epoch 430/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.3853e-05 - val_loss: 0.0504
Epoch 431/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.3540e-05 - val_loss: 0.0505
Epoch 432/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.3353e-05 - val_loss: 0.0505
Epoch 433/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - loss: 2.3099e-05 - val_loss: 0.050
5
Epoch 434/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.2865e-05 - val_loss: 0.0505
Epoch 435/1000
```

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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.2746e-05 - val_loss: 0.0506
Epoch 436/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.2402e-05 - val_loss: 0.0507
Epoch 437/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.2219e-05 - val_loss: 0.0507
Epoch 438/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.1964e-05 - val_loss: 0.0508
Epoch 439/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.1840e-05 - val_loss: 0.0508
Epoch 440/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.1606e-05 - val_loss: 0.0509
Epoch 441/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.1332e-05 - val_loss: 0.0510
Epoch 442/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.1197e-05 - val_loss: 0.0510
Epoch 443/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0942e-05 - val_loss: 0.0511
Epoch 444/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0776e-05 - val_loss: 0.0511
Epoch 445/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0569e-05 - val_loss: 0.0511
Epoch 446/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0337e-05 - val_loss: 0.0512
Epoch 447/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0202e-05 - val_loss: 0.0512
Epoch 448/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9958e-05 - val_loss: 0.0513
Epoch 449/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9838e-05 - val_loss: 0.0513
Epoch 450/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 1.9551e-05 - val_loss: 0.051
3
Epoch 451/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9399e-05 - val_loss: 0.0513
Epoch 452/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.9195e-05 - val_loss: 0.0515
Epoch 453/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9050e-05 - val_loss: 0.0515
Epoch 454/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.8836e-05 - val_loss: 0.0515
Epoch 455/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8659e-05 - val_loss: 0.0515
Epoch 456/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8459e-05 - val_loss: 0.0516
Epoch 457/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1.8375e-05 - val_loss: 0.0516
Epoch 458/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8123e-05 - val_loss: 0.0517
Epoch 459/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7932e-05 - val_loss: 0.0518
Epoch 460/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7819e-05 - val_loss: 0.0517
Epoch 461/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7630e-05 - val_loss: 0.0519
Epoch 462/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7426e-05 - val_loss: 0.0520
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Epoch 463/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7306e-05 - val_loss: 0.0519
Epoch 464/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7098e-05 - val_loss: 0.0520
Epoch 465/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6965e-05 - val_loss: 0.0520
Epoch 466/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6808e-05 - val_loss: 0.0521
Epoch 467/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6619e-05 - val_loss: 0.0521
Epoch 468/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6487e-05 - val_loss: 0.0522
Epoch 469/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6357e-05 - val_loss: 0.0522
Epoch 470/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6218e-05 - val_loss: 0.0521
Epoch 471/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6008e-05 - val_loss: 0.0523
Epoch 472/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5879e-05 - val_loss: 0.0523
Epoch 473/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5697e-05 - val_loss: 0.0524
Epoch 474/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5587e-05 - val_loss: 0.0524
Epoch 475/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5438e-05 - val_loss: 0.0524
Epoch 476/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5243e-05 - val_loss: 0.0525
Epoch 477/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.5158e-05 - val_loss: 0.0525
Epoch 478/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5014e-05 - val_loss: 0.0526
Epoch 479/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.4915e-05 - val_loss: 0.0525
Epoch 480/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 1.4697e-05 - val_loss: 0.0527
Epoch 481/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.4588e-05 - val_loss: 0.0527
Epoch 482/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - loss: 1.4419e-05 - val_loss: 0.052
8
Epoch 483/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4315e-05 - val_loss: 0.0527
Epoch 484/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.4177e-05 - val_loss: 0.0528
Epoch 485/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4031e-05 - val_loss: 0.0529
Epoch 486/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.3935e-05 - val_loss: 0.0529
Epoch 487/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3775e-05 - val_loss: 0.0530
Epoch 488/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3700e-05 - val_loss: 0.0530
Epoch 489/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.3509e-05 - val_loss: 0.0531
Epoch 490/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3414e-05 - val_loss: 0.0531
Epoch 491/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3257e-05 - val_loss: 0.0532
Epoch 492/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3156e-05 - val_loss: 0.0532
Epoch 493/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3030e-05 - val_loss: 0.0533
Epoch 494/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2881e-05 - val_loss: 0.0533
Epoch 495/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2850e-05 - val_loss: 0.0532
Epoch 496/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2685e-05 - val_loss: 0.0533
Epoch 497/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.2534e-05 - val_loss: 0.0534
Epoch 498/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2440e-05 - val_loss: 0.0534
Epoch 499/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.2319e-05 - val_loss: 0.0536
Epoch 500/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2207e-05 - val_loss: 0.0536
Epoch 501/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2116e-05 - val_loss: 0.0536
Epoch 502/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1968e-05 - val_loss: 0.0537
Epoch 503/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1884e-05 - val_loss: 0.0538
Epoch 504/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.1761e-05 - val_loss: 0.0538
Epoch 505/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1637e-05 - val_loss: 0.0538
Epoch 506/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1596e-05 - val_loss: 0.0538
Epoch 507/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.1456e-05 - val_loss: 0.0539
Epoch 508/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1328e-05 - val_loss: 0.0539
Epoch 509/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.1203e-05 - val_loss: 0.054
0
Epoch 510/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 1.1143e-05 - val_loss: 0.054
0
Epoch 511/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 1.1015e-05 - val_loss: 0.054
0
Epoch 512/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.0920e-05 - val_loss: 0.0540
Epoch 513/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0812e-05 - val_loss: 0.0541
Epoch 514/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0700e-05 - val_loss: 0.0542
Epoch 515/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0616e-05 - val_loss: 0.0542
Epoch 516/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.0512e-05 - val_loss: 0.0543
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Epoch 517/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.0441e-05 - val_loss: 0.0542
Epoch 518/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0303e-05 - val_loss: 0.0544
Epoch 519/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0224e-05 - val_loss: 0.0544
Epoch 520/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0118e-05 - val_loss: 0.0544
Epoch 521/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0024e-05 - val_loss: 0.0544
Epoch 522/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.9398e-06 - val_loss: 0.0545
Epoch 523/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 9.8354e-06 - val_loss: 0.0546
Epoch 524/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.7635e-06 - val_loss: 0.0546
Epoch 525/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 9.6580e-06 - val_loss: 0.0547
Epoch 526/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.5722e-06 - val_loss: 0.0547
Epoch 527/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.4810e-06 - val_loss: 0.0547
Epoch 528/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.3811e-06 - val_loss: 0.0547
Epoch 529/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 9.3497e-06 - val_loss: 0.0547
Epoch 530/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.2318e-06 - val_loss: 0.0548
Epoch 531/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 9.1219e-06 - val_loss: 0.054
8
Epoch 532/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.0400e-06 - val_loss: 0.0549
Epoch 533/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 8.9936e-06 - val_loss: 0.0549
Epoch 534/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 8.8813e-06 - val_loss: 0.0550
Epoch 535/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.7980e-06 - val_loss: 0.0550
Epoch 536/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.7173e-06 - val_loss: 0.0551
Epoch 537/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.6418e-06 - val_loss: 0.0551
Epoch 538/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.5404e-06 - val_loss: 0.0552
Epoch 539/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.4928e-06 - val_loss: 0.0551
Epoch 540/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.4112e-06 - val_loss: 0.0552
Epoch 541/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 8.3279e-06 - val_loss: 0.055
2
Epoch 542/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.2479e-06 - val_loss: 0.0553
Epoch 543/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.1876e-06 - val_loss: 0.0554
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Epoch 544/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.1073e-06 - val_loss: 0.0554
Epoch 545/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 8.0183e-06 - val_loss: 0.0554
Epoch 546/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.9526e-06 - val_loss: 0.0554
Epoch 547/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 7.8807e-06 - val_loss: 0.0555
Epoch 548/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.8321e-06 - val_loss: 0.0555
Epoch 549/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.7295e-06 - val_loss: 0.0556
Epoch 550/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 7.6679e-06 - val_loss: 0.0556
Epoch 551/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.5866e-06 - val_loss: 0.0557
Epoch 552/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.5145e-06 - val_loss: 0.0556
Epoch 553/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.4550e-06 - val_loss: 0.0558
Epoch 554/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.3842e-06 - val_loss: 0.0558
Epoch 555/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.3037e-06 - val_loss: 0.0558
Epoch 556/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.2523e-06 - val_loss: 0.0558
Epoch 557/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.1884e-06 - val_loss: 0.0559
Epoch 558/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.1125e-06 - val_loss: 0.0559
Epoch 559/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.0477e-06 - val_loss: 0.0560
Epoch 560/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 6.9832e-06 - val_loss: 0.0560
Epoch 561/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 6.9503e-06 - val_loss: 0.0560
Epoch 562/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 6.8531e-06 - val_loss: 0.0560
Epoch 563/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.7812e-06 - val_loss: 0.0561
Epoch 564/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.7235e-06 - val_loss: 0.0562
Epoch 565/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 6.6673e-06 - val_loss: 0.0562
Epoch 566/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.6060e-06 - val_loss: 0.0561
Epoch 567/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.5341e-06 - val_loss: 0.0563
Epoch 568/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.4987e-06 - val_loss: 0.0563
Epoch 569/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.4212e-06 - val_loss: 0.0563
Epoch 570/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 6.3666e-06 - val_loss: 0.0564
Epoch 571/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 6.3177e-06 - val_loss: 0.056
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4
Epoch 572/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.2499e-06 - val_loss: 0.0565
Epoch 573/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 6.1973e-06 - val_loss: 0.0564
Epoch 574/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.1488e-06 - val_loss: 0.0566
Epoch 575/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 6.0793e-06 - val_loss: 0.0566
Epoch 576/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.0481e-06 - val_loss: 0.0565
Epoch 577/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.9716e-06 - val_loss: 0.0566
Epoch 578/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 5.9191e-06 - val_loss: 0.0567
Epoch 579/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.8527e-06 - val_loss: 0.0567
Epoch 580/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 5.8163e-06 - val_loss: 0.0568
Epoch 581/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.7573e-06 - val_loss: 0.0568
Epoch 582/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.7039e-06 - val_loss: 0.0568
Epoch 583/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.6609e-06 - val_loss: 0.0569
Epoch 584/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 5.6081e-06 - val_loss: 0.057
0
Epoch 585/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.5555e-06 - val_loss: 0.0569
Epoch 586/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.5073e-06 - val_loss: 0.0569
Epoch 587/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.4476e-06 - val_loss: 0.0571
Epoch 588/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 5.4086e-06 - val_loss: 0.0571
Epoch 589/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 5.3535e-06 - val_loss: 0.0571
Epoch 590/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.3065e-06 - val_loss: 0.0571
Epoch 591/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.2712e-06 - val_loss: 0.0572
Epoch 592/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.2098e-06 - val_loss: 0.0572
Epoch 593/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.1890e-06 - val_loss: 0.0572
Epoch 594/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.1095e-06 - val_loss: 0.0573
Epoch 595/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 5.0833e-06 - val_loss: 0.0573
Epoch 596/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 5.0267e-06 - val_loss: 0.0574
Epoch 597/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 4.9854e-06 - val_loss: 0.0574
Epoch 598/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 4.9357e-06 - val_loss: 0.0574
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Epoch 599/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 4.9009e-06 - val_loss: 0.057
4
Epoch 600/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.8524e-06 - val_loss: 0.0576
Epoch 601/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.8059e-06 - val_loss: 0.0576
Epoch 602/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.7651e-06 - val_loss: 0.0576
Epoch 603/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.7261e-06 - val_loss: 0.0577
Epoch 604/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.6765e-06 - val_loss: 0.0577
Epoch 605/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.6619e-06 - val_loss: 0.0576
Epoch 606/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.5923e-06 - val_loss: 0.0577
Epoch 607/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 4.5525e-06 - val_loss: 0.057
8
Epoch 608/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.5110e-06 - val_loss: 0.0577
Epoch 609/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.4728e-06 - val_loss: 0.0579
Epoch 610/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.4410e-06 - val_loss: 0.0579
Epoch 611/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.3914e-06 - val_loss: 0.0579
Epoch 612/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.3593e-06 - val_loss: 0.0579
Epoch 613/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.3200e-06 - val_loss: 0.0580
Epoch 614/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.2780e-06 - val_loss: 0.0580
Epoch 615/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.2442e-06 - val_loss: 0.0581
Epoch 616/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.1983e-06 - val_loss: 0.0581
Epoch 617/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 4.1757e-06 - val_loss: 0.058
1
Epoch 618/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.1309e-06 - val_loss: 0.0582
Epoch 619/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.0962e-06 - val_loss: 0.0582
Epoch 620/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.0590e-06 - val_loss: 0.0583
Epoch 621/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 4.0192e-06 - val_loss: 0.0583
Epoch 622/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.9919e-06 - val_loss: 0.0583
Epoch 623/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.9560e-06 - val_loss: 0.0584
Epoch 624/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.9182e-06 - val_loss: 0.0584
Epoch 625/1000
```

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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.8882e-06 - val_loss: 0.0584
Epoch 626/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 3.8443e-06 - val_loss: 0.058
4
Epoch 627/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.8184e-06 - val_loss: 0.0585
Epoch 628/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.7807e-06 - val_loss: 0.0586
Epoch 629/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 3.7459e-06 - val_loss: 0.058
5
Epoch 630/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.7167e-06 - val_loss: 0.0586
Epoch 631/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.6841e-06 - val_loss: 0.0587
Epoch 632/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.6497e-06 - val_loss: 0.0587
Epoch 633/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 3.6288e-06 - val_loss: 0.0587
Epoch 634/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 3.5924e-06 - val_loss: 0.058
8
Epoch 635/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.5555e-06 - val_loss: 0.0588
Epoch 636/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.5125e-06 - val_loss: 0.0589
Epoch 637/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 3.4978e-06 - val_loss: 0.058
9
Epoch 638/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.4639e-06 - val_loss: 0.0589
Epoch 639/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.4365e-06 - val_loss: 0.0589
Epoch 640/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.4083e-06 - val_loss: 0.0590
Epoch 641/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.3688e-06 - val_loss: 0.0591
Epoch 642/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.3529e-06 - val_loss: 0.0590
Epoch 643/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.3167e-06 - val_loss: 0.0591
Epoch 644/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.2843e-06 - val_loss: 0.0591
Epoch 645/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.2555e-06 - val_loss: 0.0591
Epoch 646/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.2324e-06 - val_loss: 0.0592
Epoch 647/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.1947e-06 - val_loss: 0.0593
Epoch 648/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.1773e-06 - val_loss: 0.0593
Epoch 649/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.1448e-06 - val_loss: 0.0594
Epoch 650/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.1223e-06 - val_loss: 0.0594
Epoch 651/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.0915e-06 - val_loss: 0.0594
Epoch 652/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.0613e-06 - val_loss: 0.0595
Epoch 653/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.0409e-06 - val_loss: 0.0595
Epoch 654/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.0116e-06 - val_loss: 0.0596
Epoch 655/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.9848e-06 - val_loss: 0.0595
Epoch 656/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.9654e-06 - val_loss: 0.0596
Epoch 657/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.9295e-06 - val_loss: 0.0597
Epoch 658/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.9227e-06 - val_loss: 0.0597
Epoch 659/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.8792e-06 - val_loss: 0.0597
Epoch 660/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.8612e-06 - val_loss: 0.0597
Epoch 661/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 2.8319e-06 - val_loss: 0.059
8
Epoch 662/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.8098e-06 - val_loss: 0.0599
Epoch 663/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7899e-06 - val_loss: 0.0599
Epoch 664/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7639e-06 - val_loss: 0.0600
Epoch 665/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7381e-06 - val_loss: 0.0600
Epoch 666/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7128e-06 - val_loss: 0.0601
Epoch 667/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.6915e-06 - val_loss: 0.0601
Epoch 668/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.6673e-06 - val_loss: 0.0601
Epoch 669/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 2.6543e-06 - val_loss: 0.0601
Epoch 670/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.6221e-06 - val_loss: 0.0602
Epoch 671/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.6061e-06 - val_loss: 0.0602
Epoch 672/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.5755e-06 - val_loss: 0.0603
Epoch 673/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.5600e-06 - val_loss: 0.0603
Epoch 674/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 2.5326e-06 - val_loss: 0.0604
Epoch 675/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 2.5132e-06 - val_loss: 0.0604
Epoch 676/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 2.4937e-06 - val_loss: 0.0605
Epoch 677/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.4690e-06 - val_loss: 0.0606
Epoch 678/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 2.4506e-06 - val_loss: 0.060
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Epoch 679/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 7ms/step - loss: 2.4290e-06 - val_loss: 0.0606
Epoch 680/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 8ms/step - loss: 2.4194e-06 - val_loss: 0.0606
Epoch 681/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.3847e-06 - val_loss: 0.0606
Epoch 682/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.3712e-06 - val_loss: 0.0607
Epoch 683/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 5ms/step - loss: 2.3457e-06 - val_loss: 0.0608
Epoch 684/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 9ms/step - loss: 2.3280e-06 - val_loss: 0.0607
Epoch 685/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.3068e-06 - val_loss: 0.0609
Epoch 686/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.2902e-06 - val_loss: 0.0609
Epoch 687/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 8ms/step - loss: 2.2731e-06 - val_loss: 0.0609
Epoch 688/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 19ms/step - loss: 2.2464e-06 - val_loss: 0.060
9
Epoch 689/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.2350e-06 - val_loss: 0.0610
Epoch 690/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.2112e-06 - val_loss: 0.0610
Epoch 691/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.1949e-06 - val_loss: 0.0610
Epoch 692/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.1758e-06 - val_loss: 0.0612
Epoch 693/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 9ms/step - loss: 2.1528e-06 - val_loss: 0.0612
Epoch 694/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.1463e-06 - val_loss: 0.0612
Epoch 695/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.1225e-06 - val_loss: 0.0613
Epoch 696/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.1008e-06 - val_loss: 0.0613
Epoch 697/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.0846e-06 - val_loss: 0.0614
Epoch 698/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.0644e-06 - val_loss: 0.0614
Epoch 699/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.0477e-06 - val_loss: 0.0614
Epoch 700/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.0341e-06 - val_loss: 0.0614
Epoch 701/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 2.0167e-06 - val_loss: 0.0615
Epoch 702/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 7ms/step - loss: 2.0008e-06 - val_loss: 0.0616
Epoch 703/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 10ms/step - loss: 1.9809e-06 - val_loss: 0.061
6
Epoch 704/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ **0s** 7ms/step - loss: 1.9638e-06 - val_loss: 0.0617
Epoch 705/1000

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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.9545e-06 - val_loss: 0.0617
Epoch 706/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9273e-06 - val_loss: 0.0618
Epoch 707/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 1.9184e-06 - val_loss: 0.0618
Epoch 708/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8989e-06 - val_loss: 0.0618
Epoch 709/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8845e-06 - val_loss: 0.0618
Epoch 710/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8668e-06 - val_loss: 0.0619
Epoch 711/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8538e-06 - val_loss: 0.0619
Epoch 712/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8347e-06 - val_loss: 0.0620
Epoch 713/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.8219e-06 - val_loss: 0.0620
Epoch 714/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8069e-06 - val_loss: 0.0620
Epoch 715/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7878e-06 - val_loss: 0.0621
Epoch 716/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7821e-06 - val_loss: 0.0621
Epoch 717/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7595e-06 - val_loss: 0.0622
Epoch 718/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7396e-06 - val_loss: 0.0623
Epoch 719/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.7352e-06 - val_loss: 0.0622
Epoch 720/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.7188e-06 - val_loss: 0.0623
Epoch 721/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7025e-06 - val_loss: 0.0624
Epoch 722/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 1.6891e-06 - val_loss: 0.062
4
Epoch 723/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.6749e-06 - val_loss: 0.0625
Epoch 724/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6573e-06 - val_loss: 0.0625
Epoch 725/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 1.6522e-06 - val_loss: 0.0625
Epoch 726/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6329e-06 - val_loss: 0.0625
Epoch 727/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6156e-06 - val_loss: 0.0626
Epoch 728/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.6012e-06 - val_loss: 0.0626
Epoch 729/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5922e-06 - val_loss: 0.0627
Epoch 730/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 1.5786e-06 - val_loss: 0.062
8
Epoch 731/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5661e-06 - val_loss: 0.0628
Epoch 732/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5516e-06 - val_loss: 0.0628
Epoch 733/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5378e-06 - val_loss: 0.0629
Epoch 734/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5312e-06 - val_loss: 0.0629
Epoch 735/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.5157e-06 - val_loss: 0.0630
Epoch 736/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4988e-06 - val_loss: 0.0630
Epoch 737/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4895e-06 - val_loss: 0.0630
Epoch 738/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.4770e-06 - val_loss: 0.0631
Epoch 739/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4654e-06 - val_loss: 0.0631
Epoch 740/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4532e-06 - val_loss: 0.0632
Epoch 741/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4388e-06 - val_loss: 0.0632
Epoch 742/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4288e-06 - val_loss: 0.0633
Epoch 743/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4153e-06 - val_loss: 0.0633
Epoch 744/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.4019e-06 - val_loss: 0.0633
Epoch 745/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3954e-06 - val_loss: 0.0634
Epoch 746/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3807e-06 - val_loss: 0.0634
Epoch 747/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1.3700e-06 - val_loss: 0.0635
Epoch 748/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.3593e-06 - val_loss: 0.0636
Epoch 749/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3476e-06 - val_loss: 0.0636
Epoch 750/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.3334e-06 - val_loss: 0.0636
Epoch 751/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3254e-06 - val_loss: 0.0636
Epoch 752/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3131e-06 - val_loss: 0.0637
Epoch 753/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.3039e-06 - val_loss: 0.0638
Epoch 754/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2942e-06 - val_loss: 0.0638
Epoch 755/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2812e-06 - val_loss: 0.0639
Epoch 756/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2736e-06 - val_loss: 0.0639
Epoch 757/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2614e-06 - val_loss: 0.0639
Epoch 758/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.2483e-06 - val_loss: 0.0640
Epoch 759/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2412e-06 - val_loss: 0.0640
Epoch 760/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2310e-06 - val_loss: 0.0640
Epoch 761/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2180e-06 - val_loss: 0.0641
Epoch 762/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.2099e-06 - val_loss: 0.0642
Epoch 763/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1982e-06 - val_loss: 0.0642
Epoch 764/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1900e-06 - val_loss: 0.0642
Epoch 765/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1813e-06 - val_loss: 0.0643
Epoch 766/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1.1687e-06 - val_loss: 0.0643
Epoch 767/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1637e-06 - val_loss: 0.0643
Epoch 768/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1522e-06 - val_loss: 0.0644
Epoch 769/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1394e-06 - val_loss: 0.0645
Epoch 770/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 1.1324e-06 - val_loss: 0.064
5
Epoch 771/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.1244e-06 - val_loss: 0.0646
Epoch 772/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.1111e-06 - val_loss: 0.0646
Epoch 773/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1.1042e-06 - val_loss: 0.0646
Epoch 774/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 1.0946e-06 - val_loss: 0.064
7
Epoch 775/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 1.0914e-06 - val_loss: 0.0646
Epoch 776/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0757e-06 - val_loss: 0.0647
Epoch 777/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.0691e-06 - val_loss: 0.064
8
Epoch 778/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0584e-06 - val_loss: 0.0648
Epoch 779/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0515e-06 - val_loss: 0.0648
Epoch 780/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0428e-06 - val_loss: 0.0649
Epoch 781/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0339e-06 - val_loss: 0.0649
Epoch 782/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0282e-06 - val_loss: 0.0649
Epoch 783/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0179e-06 - val_loss: 0.0650
Epoch 784/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0074e-06 - val_loss: 0.0650
Epoch 785/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.0003e-06 - val_loss: 0.0650
Epoch 786/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.9151e-07 - val_loss: 0.0651
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Epoch 787/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 9.8290e-07 - val_loss: 0.0652
Epoch 788/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 9.7679e-07 - val_loss: 0.0652
Epoch 789/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.6826e-07 - val_loss: 0.0652
Epoch 790/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.5961e-07 - val_loss: 0.0653
Epoch 791/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.5103e-07 - val_loss: 0.0653
Epoch 792/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 9.4773e-07 - val_loss: 0.0653
Epoch 793/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.3632e-07 - val_loss: 0.0653
Epoch 794/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 9.2600e-07 - val_loss: 0.0654
Epoch 795/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 9.2111e-07 - val_loss: 0.0655
Epoch 796/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 9.1208e-07 - val_loss: 0.0655
Epoch 797/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.0681e-07 - val_loss: 0.0655
Epoch 798/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.9936e-07 - val_loss: 0.0655
Epoch 799/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 8.9205e-07 - val_loss: 0.0656
Epoch 800/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.8934e-07 - val_loss: 0.0655
Epoch 801/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.7945e-07 - val_loss: 0.0657
Epoch 802/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.6850e-07 - val_loss: 0.0657
Epoch 803/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.6240e-07 - val_loss: 0.0657
Epoch 804/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.5457e-07 - val_loss: 0.0657
Epoch 805/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.4953e-07 - val_loss: 0.0658
Epoch 806/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.4232e-07 - val_loss: 0.0659
Epoch 807/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.3565e-07 - val_loss: 0.0659
Epoch 808/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 8.2823e-07 - val_loss: 0.0660
Epoch 809/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.2070e-07 - val_loss: 0.0660
Epoch 810/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.1823e-07 - val_loss: 0.0659
Epoch 811/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 8.0792e-07 - val_loss: 0.0661
Epoch 812/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 8.0183e-07 - val_loss: 0.0661
Epoch 813/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.9529e-07 - val_loss: 0.0661
Epoch 814/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 7.8828e-07 - val_loss: 0.0662
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Epoch 815/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 7.8137e-07 - val_loss: 0.0662
Epoch 816/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 7.7491e-07 - val_loss: 0.0662
Epoch 817/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.7067e-07 - val_loss: 0.0663
Epoch 818/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.6329e-07 - val_loss: 0.0663
Epoch 819/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.5916e-07 - val_loss: 0.0663
Epoch 820/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.5146e-07 - val_loss: 0.0664
Epoch 821/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 7.4449e-07 - val_loss: 0.0664
Epoch 822/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.3866e-07 - val_loss: 0.0664
Epoch 823/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 7.3465e-07 - val_loss: 0.0665
Epoch 824/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.2566e-07 - val_loss: 0.0666
Epoch 825/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.2131e-07 - val_loss: 0.0666
Epoch 826/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 7.1558e-07 - val_loss: 0.0666
Epoch 827/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 7.1090e-07 - val_loss: 0.066
6
Epoch 828/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 7.0297e-07 - val_loss: 0.0666
Epoch 829/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.9761e-07 - val_loss: 0.0667
Epoch 830/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.9054e-07 - val_loss: 0.0667
Epoch 831/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.8658e-07 - val_loss: 0.0667
Epoch 832/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.8157e-07 - val_loss: 0.0669
Epoch 833/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.7683e-07 - val_loss: 0.0669
Epoch 834/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 6.7030e-07 - val_loss: 0.0669
Epoch 835/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.6383e-07 - val_loss: 0.0669
Epoch 836/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.6000e-07 - val_loss: 0.0670
Epoch 837/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 6.5369e-07 - val_loss: 0.0671
Epoch 838/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 6.5073e-07 - val_loss: 0.0671
Epoch 839/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.4315e-07 - val_loss: 0.0671
Epoch 840/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 6.3833e-07 - val_loss: 0.0671
Epoch 841/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.3314e-07 - val_loss: 0.0672
Epoch 842/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.2777e-07 - val_loss: 0.0673
Epoch 843/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.2315e-07 - val_loss: 0.0672
Epoch 844/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 6.1658e-07 - val_loss: 0.067
3
Epoch 845/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.1509e-07 - val_loss: 0.0673
Epoch 846/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.0804e-07 - val_loss: 0.0673
Epoch 847/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 6.0322e-07 - val_loss: 0.0675
Epoch 848/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.9781e-07 - val_loss: 0.0675
Epoch 849/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.9314e-07 - val_loss: 0.0675
Epoch 850/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 5.8796e-07 - val_loss: 0.067
6
Epoch 851/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 5.8362e-07 - val_loss: 0.0676
Epoch 852/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 5.7852e-07 - val_loss: 0.0676
Epoch 853/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.7391e-07 - val_loss: 0.0677
Epoch 854/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.6959e-07 - val_loss: 0.0677
Epoch 855/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 5.6764e-07 - val_loss: 0.067
7
Epoch 856/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 5.5993e-07 - val_loss: 0.0678
Epoch 857/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 5.5564e-07 - val_loss: 0.0678
Epoch 858/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.5124e-07 - val_loss: 0.0679
Epoch 859/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.4512e-07 - val_loss: 0.0679
Epoch 860/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.4253e-07 - val_loss: 0.0680
Epoch 861/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.3765e-07 - val_loss: 0.0680
Epoch 862/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 5.3586e-07 - val_loss: 0.0680
Epoch 863/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.2835e-07 - val_loss: 0.0681
Epoch 864/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 5.2473e-07 - val_loss: 0.0681
Epoch 865/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 5.1982e-07 - val_loss: 0.068
2
Epoch 866/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.1684e-07 - val_loss: 0.0682
Epoch 867/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.1216e-07 - val_loss: 0.0682
Epoch 868/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 5.0721e-07 - val_loss: 0.068
3
Epoch 869/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 5.0520e-07 - val_loss: 0.0683
Epoch 870/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 5.0023e-07 - val_loss: 0.0683
Epoch 871/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 4.9483e-07 - val_loss: 0.0684
Epoch 872/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.9234e-07 - val_loss: 0.0684
Epoch 873/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 4.8740e-07 - val_loss: 0.0684
Epoch 874/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.8300e-07 - val_loss: 0.0686
Epoch 875/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 4.8005e-07 - val_loss: 0.068
6
Epoch 876/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.7614e-07 - val_loss: 0.0686
Epoch 877/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.7180e-07 - val_loss: 0.0686
Epoch 878/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.6874e-07 - val_loss: 0.0688
Epoch 879/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.6344e-07 - val_loss: 0.0688
Epoch 880/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 4.6184e-07 - val_loss: 0.068
8
Epoch 881/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.5774e-07 - val_loss: 0.0688
Epoch 882/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.5338e-07 - val_loss: 0.0689
Epoch 883/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 4.4936e-07 - val_loss: 0.069
0
Epoch 884/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.4536e-07 - val_loss: 0.0690
Epoch 885/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.4205e-07 - val_loss: 0.0690
Epoch 886/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.3828e-07 - val_loss: 0.0691
Epoch 887/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 4.3521e-07 - val_loss: 0.0691
Epoch 888/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.3217e-07 - val_loss: 0.0692
Epoch 889/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.2861e-07 - val_loss: 0.0693
Epoch 890/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.2431e-07 - val_loss: 0.0693
Epoch 891/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.2253e-07 - val_loss: 0.0692
Epoch 892/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.1727e-07 - val_loss: 0.0693
Epoch 893/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.1460e-07 - val_loss: 0.0693
Epoch 894/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.1042e-07 - val_loss: 0.0694
Epoch 895/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.0791e-07 - val_loss: 0.0695
Epoch 896/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.0481e-07 - val_loss: 0.0695
Epoch 897/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 4.0081e-07 - val_loss: 0.0695
Epoch 898/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.9804e-07 - val_loss: 0.0695
Epoch 899/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.9493e-07 - val_loss: 0.0696
Epoch 900/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 3.9243e-07 - val_loss: 0.0696
Epoch 901/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.8886e-07 - val_loss: 0.0697
Epoch 902/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.8493e-07 - val_loss: 0.0697
Epoch 903/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 3.8217e-07 - val_loss: 0.069
7
Epoch 904/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.7972e-07 - val_loss: 0.0699
Epoch 905/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.7583e-07 - val_loss: 0.0699
Epoch 906/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.7319e-07 - val_loss: 0.0699
Epoch 907/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 3.7008e-07 - val_loss: 0.0700
Epoch 908/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.6734e-07 - val_loss: 0.0700
Epoch 909/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.6412e-07 - val_loss: 0.0700
Epoch 910/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.6117e-07 - val_loss: 0.0701
Epoch 911/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.5961e-07 - val_loss: 0.0700
Epoch 912/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 3.5565e-07 - val_loss: 0.0701
Epoch 913/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.5254e-07 - val_loss: 0.0702
Epoch 914/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.4962e-07 - val_loss: 0.0702
Epoch 915/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.4720e-07 - val_loss: 0.0702
Epoch 916/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.4448e-07 - val_loss: 0.0703
Epoch 917/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.4119e-07 - val_loss: 0.0703
Epoch 918/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.3892e-07 - val_loss: 0.0703
Epoch 919/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.3592e-07 - val_loss: 0.0705
Epoch 920/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.3300e-07 - val_loss: 0.0705
Epoch 921/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.3166e-07 - val_loss: 0.0704
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Epoch 922/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 3.2826e-07 - val_loss: 0.0705
Epoch 923/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.2519e-07 - val_loss: 0.0706
Epoch 924/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.2271e-07 - val_loss: 0.0706
Epoch 925/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.2009e-07 - val_loss: 0.0707
Epoch 926/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 3.1773e-07 - val_loss: 0.0707
Epoch 927/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.1509e-07 - val_loss: 0.0707
Epoch 928/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 3.1253e-07 - val_loss: 0.070
8
Epoch 929/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.1118e-07 - val_loss: 0.0708
Epoch 930/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.0763e-07 - val_loss: 0.0708
Epoch 931/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.0516e-07 - val_loss: 0.0709
Epoch 932/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.0261e-07 - val_loss: 0.0709
Epoch 933/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.9986e-07 - val_loss: 0.0709
Epoch 934/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.9737e-07 - val_loss: 0.0711
Epoch 935/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.9551e-07 - val_loss: 0.0711
Epoch 936/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.9335e-07 - val_loss: 0.0711
Epoch 937/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.9112e-07 - val_loss: 0.0711
Epoch 938/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 2.8814e-07 - val_loss: 0.0712
Epoch 939/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.8740e-07 - val_loss: 0.0712
Epoch 940/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.8361e-07 - val_loss: 0.0713
Epoch 941/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.8146e-07 - val_loss: 0.0713
Epoch 942/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 2.7907e-07 - val_loss: 0.0714
Epoch 943/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7714e-07 - val_loss: 0.0714
Epoch 944/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7484e-07 - val_loss: 0.0714
Epoch 945/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7237e-07 - val_loss: 0.0714
Epoch 946/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.7094e-07 - val_loss: 0.0716
Epoch 947/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 2.6836e-07 - val_loss: 0.0716
Epoch 948/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.6643e-07 - val_loss: 0.0716
Epoch 949/1000
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14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 2.6427e-07 - val_loss: 0.0717
Epoch 950/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.6318e-07 - val_loss: 0.0716
Epoch 951/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.5956e-07 - val_loss: 0.0716
Epoch 952/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.5791e-07 - val_loss: 0.0717
Epoch 953/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.5567e-07 - val_loss: 0.0717
Epoch 954/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.5400e-07 - val_loss: 0.0717
Epoch 955/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.5222e-07 - val_loss: 0.0719
Epoch 956/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.4968e-07 - val_loss: 0.0719
Epoch 957/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.4776e-07 - val_loss: 0.0719
Epoch 958/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.4647e-07 - val_loss: 0.0720
Epoch 959/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 2.4404e-07 - val_loss: 0.0720
Epoch 960/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.4282e-07 - val_loss: 0.0720
Epoch 961/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.4054e-07 - val_loss: 0.0721
Epoch 962/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.3827e-07 - val_loss: 0.0722
Epoch 963/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 2.3690e-07 - val_loss: 0.0722
Epoch 964/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.3455e-07 - val_loss: 0.0723
Epoch 965/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.3263e-07 - val_loss: 0.0724
Epoch 966/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.3101e-07 - val_loss: 0.0723
Epoch 967/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.2887e-07 - val_loss: 0.0723
Epoch 968/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.2818e-07 - val_loss: 0.0723
Epoch 969/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.2496e-07 - val_loss: 0.0724
Epoch 970/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 2.2409e-07 - val_loss: 0.072
5
Epoch 971/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.2188e-07 - val_loss: 0.0725
Epoch 972/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 2.2079e-07 - val_loss: 0.0725
Epoch 973/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 2.1832e-07 - val_loss: 0.0726
Epoch 974/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.1703e-07 - val_loss: 0.0727
Epoch 975/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.1520e-07 - val_loss: 0.0726
Epoch 976/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.1303e-07 - val_loss: 0.0728
```

```
Epoch 977/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.1215e-07 - val_loss: 0.0727
Epoch 978/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0983e-07 - val_loss: 0.0727
Epoch 979/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 2.0793e-07 - val_loss: 0.0728
Epoch 980/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0681e-07 - val_loss: 0.0729
Epoch 981/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0478e-07 - val_loss: 0.0729
Epoch 982/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0294e-07 - val_loss: 0.0730
Epoch 983/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 2.0161e-07 - val_loss: 0.0731
Epoch 984/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9978e-07 - val_loss: 0.0731
Epoch 985/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9843e-07 - val_loss: 0.0731
Epoch 986/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9650e-07 - val_loss: 0.0732
Epoch 987/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9481e-07 - val_loss: 0.0731
Epoch 988/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9321e-07 - val_loss: 0.0732
Epoch 989/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9218e-07 - val_loss: 0.0732
Epoch 990/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.9103e-07 - val_loss: 0.0732
Epoch 991/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8868e-07 - val_loss: 0.0734
Epoch 992/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8741e-07 - val_loss: 0.0734
Epoch 993/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1.8559e-07 - val_loss: 0.0734
Epoch 994/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 1.8473e-07 - val_loss: 0.073
5
Epoch 995/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8295e-07 - val_loss: 0.0736
Epoch 996/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.8103e-07 - val_loss: 0.0735
Epoch 997/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.8017e-07 - val_loss: 0.0736
Epoch 998/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7837e-07 - val_loss: 0.0736
Epoch 999/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7755e-07 - val_loss: 0.0737
Epoch 1000/1000
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1.7577e-07 - val_loss: 0.0737
```

Out[ ]: <keras.src.callbacks.history.History at 0x7fc53a9634d0>

In [11]:
```
# Get information about the model
model.summary()

# Save Training and Validation Loss in a Pandas DataFrame
```

```
model_loss = pd.DataFrame(model.history.history)
model_loss.head()
```

Model: "sequential"

Layer (type)	Output Shape	Par
dense (Dense)	(32, 30)	
dense_1 (Dense)	(32, 15)	
dense_2 (Dense)	(32, 1)	

Total params: 4,235 (16.55 KB)
Trainable params: 1,411 (5.51 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 2,824 (11.04 KB)

Out[11]:

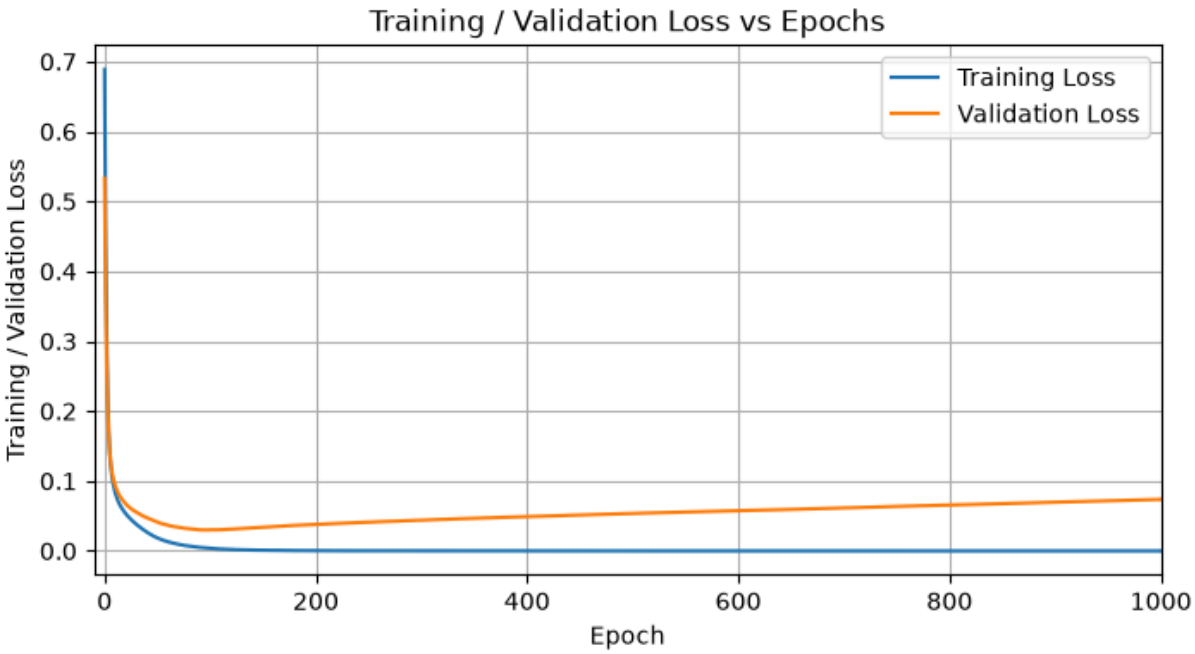
	loss	val_loss
0	0.688664	0.533090
1	0.449340	0.371635
2	0.304793	0.267862
3	0.217865	0.202885
4	0.167743	0.163855

In [12]:

```
# Plot Training / Validation Loss vs Epochs
plt.figure(figsize = (8, 4))

plt.plot(range(len(model_loss)), model_loss['loss'], label = 'Training Loss'
plt.plot(range(len(model_loss)), model_loss['val_loss'], label = 'Validatior

plt.xlim([-10, len(model_loss)])
plt.xlabel('Epoch')
plt.ylabel('Training / Validation Loss')
plt.title('Training / Validation Loss vs Epochs')
plt.legend()
plt.grid()
```



Choosing too many epochs and overfitting!

Example Two: Early Stopping

We obviously trained too much! Let's use early stopping to track the val_loss and stop training once it begins increasing too much!

In [13]:
```
# Train again a new model
model = Sequential()
model.add(Dense(units=30,activation='relu'))
model.add(Dense(units=15,activation='relu'))
model.add(Dense(units=1,activation='sigmoid'))

model.compile(loss='binary_crossentropy', optimizer='adam')
```

Stop training when a monitored quantity has stopped improving.

```
    Arguments:
        monitor: Quantity to be monitored.
        min_delta: Minimum change in the monitored quantity
            to qualify as an improvement, i.e. an absolute
            change of less than min_delta, will count as no
            improvement.
        patience: Number of epochs with no improvement
            after which training will be stopped.
        verbose: verbosity mode.
        mode: One of `{"auto", "min", "max"}`. In `min` mode,
            training will stop when the quantity
            monitored has stopped decreasing; in `max`
            mode it will stop when the quantity
            monitored has stopped increasing; in `auto`
```

mode, the direction is automatically inferred
from the name of the monitored quantity.

In [14]:
```
from tensorflow.keras.callbacks import EarlyStopping                    # Impor

# Add early stopping
early_stop = EarlyStopping(monitor='val_loss', mode='min', verbose=1, patier

# Fit with early stopping
model.fit(x=X_train, y=y_train, epochs=600, validation_data=(X_test, y_test)

# Save results
model_loss = pd.DataFrame(model.history.history)            # Save Training a
```

```
Epoch 1/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 34ms/step - loss: 0.7674 - val_loss: 0.5789
Epoch 2/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.5309 - val_loss: 0.4242
Epoch 3/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.3762 - val_loss: 0.3148
Epoch 4/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2704 - val_loss: 0.2390
Epoch 5/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.2011 - val_loss: 0.1891
Epoch 6/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1586 - val_loss: 0.1569
Epoch 7/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.1320 - val_loss: 0.1349
Epoch 8/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1141 - val_loss: 0.1190
Epoch 9/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1013 - val_loss: 0.1075
Epoch 10/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0918 - val_loss: 0.0992
Epoch 11/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0843 - val_loss: 0.0930
Epoch 12/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0782 - val_loss: 0.0883
Epoch 13/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0732 - val_loss: 0.0847
Epoch 14/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0690 - val_loss: 0.0817
Epoch 15/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0654 - val_loss: 0.0793
Epoch 16/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0624 - val_loss: 0.0772
Epoch 17/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0599 - val_loss: 0.0752
Epoch 18/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0576 - val_loss: 0.0733
Epoch 19/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0557 - val_loss: 0.0719
Epoch 20/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0539 - val_loss: 0.0708
Epoch 21/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0522 - val_loss: 0.0701
Epoch 22/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0504 - val_loss: 0.0696
Epoch 23/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0488 - val_loss: 0.0689
Epoch 24/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0472 - val_loss: 0.0679
Epoch 25/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0457 - val_loss: 0.0675
Epoch 26/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0443 - val_loss: 0.0670
Epoch 27/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0429 - val_loss: 0.0666
Epoch 28/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0416 - val_loss: 0.0660
```

```
Epoch 29/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0403 - val_loss: 0.0655
Epoch 30/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0391 - val_loss: 0.0649
Epoch 31/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0379 - val_loss: 0.0644
Epoch 32/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0368 - val_loss: 0.0632
Epoch 33/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0357 - val_loss: 0.0617
Epoch 34/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0346 - val_loss: 0.0605
Epoch 35/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0336 - val_loss: 0.0590
Epoch 36/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0326 - val_loss: 0.0572
Epoch 37/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0316 - val_loss: 0.0557
Epoch 38/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0307 - val_loss: 0.0543
Epoch 39/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0298 - val_loss: 0.0532
Epoch 40/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0290 - val_loss: 0.0516
Epoch 41/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0281 - val_loss: 0.0500
Epoch 42/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0274 - val_loss: 0.0489
Epoch 43/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0266 - val_loss: 0.0477
Epoch 44/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0258 - val_loss: 0.0462
Epoch 45/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0252 - val_loss: 0.0451
Epoch 46/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0245 - val_loss: 0.0443
Epoch 47/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0238 - val_loss: 0.0434
Epoch 48/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0231 - val_loss: 0.0423
Epoch 49/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0224 - val_loss: 0.0420
Epoch 50/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0218 - val_loss: 0.0414
Epoch 51/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0211 - val_loss: 0.0405
Epoch 52/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0205 - val_loss: 0.0401
Epoch 53/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0198 - val_loss: 0.0397
Epoch 54/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0192 - val_loss: 0.0392
Epoch 55/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0186 - val_loss: 0.0387
Epoch 56/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0180 - val_loss: 0.0381
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Epoch 57/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0174 - val_loss: 0.0375
Epoch 58/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0169 - val_loss: 0.0368
Epoch 59/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0165 - val_loss: 0.0358
Epoch 60/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0159 - val_loss: 0.0348
Epoch 61/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0154 - val_loss: 0.0336
Epoch 62/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0150 - val_loss: 0.0327
Epoch 63/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0145 - val_loss: 0.0320
Epoch 64/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0141 - val_loss: 0.0310
Epoch 65/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0137 - val_loss: 0.0302
Epoch 66/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0133 - val_loss: 0.0296
Epoch 67/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0129 - val_loss: 0.0288
Epoch 68/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0125 - val_loss: 0.0280
Epoch 69/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0122 - val_loss: 0.0274
Epoch 70/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0118 - val_loss: 0.0269
Epoch 71/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0115 - val_loss: 0.0260
Epoch 72/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0111 - val_loss: 0.0255
Epoch 73/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0108 - val_loss: 0.0250
Epoch 74/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0105 - val_loss: 0.0243
Epoch 75/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0102 - val_loss: 0.0238
Epoch 76/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0099 - val_loss: 0.0234
Epoch 77/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0096 - val_loss: 0.0233
Epoch 78/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0093 - val_loss: 0.0229
Epoch 79/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0091 - val_loss: 0.0223
Epoch 80/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0088 - val_loss: 0.0218
Epoch 81/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0084 - val_loss: 0.0216
Epoch 82/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0081 - val_loss: 0.0213
Epoch 83/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0079 - val_loss: 0.0210
Epoch 84/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0076 - val_loss: 0.0206
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Epoch 85/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0074 - val_loss: 0.0203
Epoch 86/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0072 - val_loss: 0.0200
Epoch 87/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0069 - val_loss: 0.0195
Epoch 88/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0068 - val_loss: 0.0194
Epoch 89/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0065 - val_loss: 0.0192
Epoch 90/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0063 - val_loss: 0.0188
Epoch 91/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0061 - val_loss: 0.0185
Epoch 92/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0060 - val_loss: 0.0184
Epoch 93/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0058 - val_loss: 0.0182
Epoch 94/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0057 - val_loss: 0.0179
Epoch 95/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0055 - val_loss: 0.0177
Epoch 96/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0053 - val_loss: 0.0176
Epoch 97/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0052 - val_loss: 0.0174
Epoch 98/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0051 - val_loss: 0.0171
Epoch 99/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0049 - val_loss: 0.0170
Epoch 100/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0048 - val_loss: 0.0167
Epoch 101/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 0.0046 - val_loss: 0.0166
Epoch 102/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0045 - val_loss: 0.0164
Epoch 103/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0044 - val_loss: 0.0162
Epoch 104/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0043 - val_loss: 0.0160
Epoch 105/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0042 - val_loss: 0.0159
Epoch 106/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0041 - val_loss: 0.0157
Epoch 107/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0040 - val_loss: 0.0156
Epoch 108/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0038 - val_loss: 0.0156
Epoch 109/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0038 - val_loss: 0.0155
Epoch 110/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0037 - val_loss: 0.0154
Epoch 111/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0036 - val_loss: 0.0153
Epoch 112/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0035 - val_loss: 0.0152
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Epoch 113/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0034 - val_loss: 0.0152
Epoch 114/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0033 - val_loss: 0.0150
Epoch 115/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0032 - val_loss: 0.0150
Epoch 116/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0031 - val_loss: 0.0151
Epoch 117/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0031 - val_loss: 0.0148
Epoch 118/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0030 - val_loss: 0.0148
Epoch 119/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0029 - val_loss: 0.0147
Epoch 120/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0029 - val_loss: 0.0147
Epoch 121/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0028 - val_loss: 0.0147
Epoch 122/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0028 - val_loss: 0.0146
Epoch 123/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0027 - val_loss: 0.0146
Epoch 124/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0026 - val_loss: 0.0146
Epoch 125/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0026 - val_loss: 0.0146
Epoch 126/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0025 - val_loss: 0.0145
Epoch 127/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0025 - val_loss: 0.0145
Epoch 128/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0024 - val_loss: 0.0145
Epoch 129/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0023 - val_loss: 0.0145
Epoch 130/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0023 - val_loss: 0.0144
Epoch 131/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0022 - val_loss: 0.0144
Epoch 132/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0022 - val_loss: 0.0144
Epoch 133/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0021 - val_loss: 0.0144
Epoch 134/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0021 - val_loss: 0.0143
Epoch 135/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0021 - val_loss: 0.0143
Epoch 136/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0020 - val_loss: 0.0143
Epoch 137/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0020 - val_loss: 0.0142
Epoch 138/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0019 - val_loss: 0.0143
Epoch 139/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0019 - val_loss: 0.0142
Epoch 140/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0019 - val_loss: 0.0142
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Epoch 141/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0018 - val_loss: 0.0143
Epoch 142/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0018 - val_loss: 0.0142
Epoch 143/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0017 - val_loss: 0.0143
Epoch 144/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0017 - val_loss: 0.0142
Epoch 145/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0017 - val_loss: 0.0141
Epoch 146/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0016 - val_loss: 0.0141
Epoch 147/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0016 - val_loss: 0.0141
Epoch 148/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0016 - val_loss: 0.0141
Epoch 149/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0015 - val_loss: 0.0141
Epoch 150/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0015 - val_loss: 0.0142
Epoch 151/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0015 - val_loss: 0.0142
Epoch 152/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0014 - val_loss: 0.0141
Epoch 153/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0014 - val_loss: 0.0141
Epoch 154/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0014 - val_loss: 0.0141
Epoch 155/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0014 - val_loss: 0.0141
Epoch 156/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0013 - val_loss: 0.0141
Epoch 157/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0013 - val_loss: 0.0142
Epoch 158/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0013 - val_loss: 0.0140
Epoch 159/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0013 - val_loss: 0.0140
Epoch 160/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.0012 - val_loss: 0.0141
Epoch 161/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0012 - val_loss: 0.0141
Epoch 162/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0012 - val_loss: 0.0141
Epoch 163/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0012 - val_loss: 0.0141
Epoch 164/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0012 - val_loss: 0.0142
Epoch 165/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0011 - val_loss: 0.0141
Epoch 166/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0011 - val_loss: 0.0141
Epoch 167/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0011 - val_loss: 0.0143
Epoch 168/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0011 - val_loss: 0.0141
```

```
Epoch 169/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0010 - val_loss: 0.0141
Epoch 170/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0010 - val_loss: 0.0142
Epoch 171/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0010 - val_loss: 0.0140
Epoch 172/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.9404e-04 - val_loss: 0.0141
Epoch 173/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 9.7491e-04 - val_loss: 0.014
3
Epoch 174/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 9.6275e-04 - val_loss: 0.0141
Epoch 175/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 9.3760e-04 - val_loss: 0.0142
Epoch 176/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 9.2377e-04 - val_loss: 0.0141
Epoch 177/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 9.0935e-04 - val_loss: 0.0141
Epoch 178/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 8.8989e-04 - val_loss: 0.0142
Epoch 179/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 8.7693e-04 - val_loss: 0.0142
Epoch 180/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 8.6169e-04 - val_loss: 0.0142
Epoch 181/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 8.4440e-04 - val_loss: 0.0142
Epoch 182/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 8.3166e-04 - val_loss: 0.0142
Epoch 183/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 8.1489e-04 - val_loss: 0.0142
Epoch 183: early stopping
```

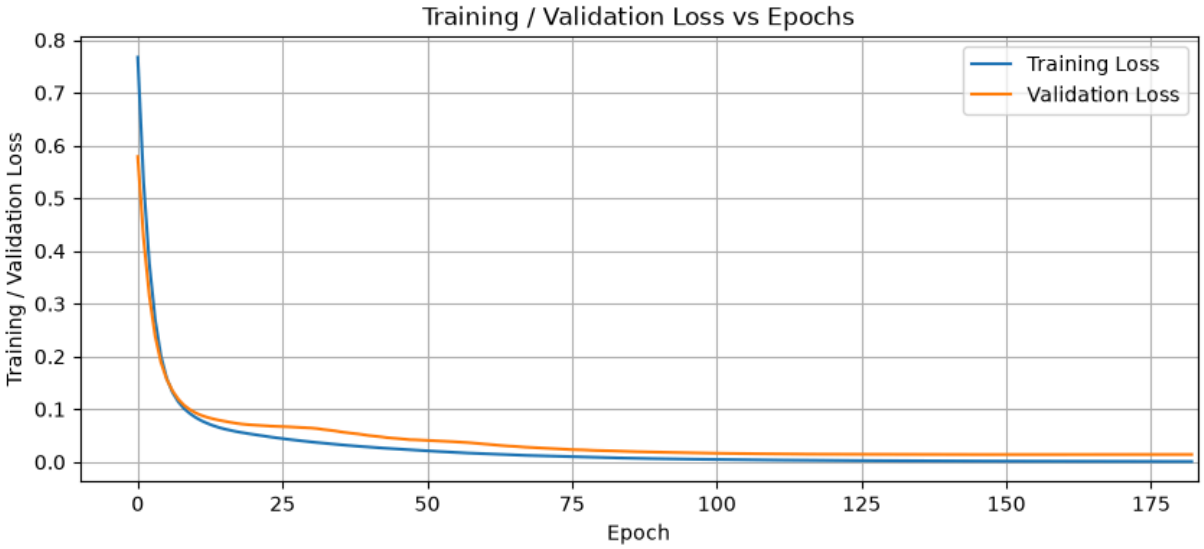
In [15]:

```python
# Plot Training / Validation Loss vs Epochs

plt.figure(figsize = (10, 4))

plt.plot(range(len(model_loss)), model_loss['loss'], label = 'Training Loss'
plt.plot(range(len(model_loss)), model_loss['val_loss'], label = 'Validatior

plt.xlim([-10, len(model_loss)])
plt.xlabel('Epoch')
plt.ylabel('Training / Validation Loss')
plt.title('Training / Validation Loss vs Epochs')
plt.legend()
plt.grid()
```



Example Three: Adding in DropOut Layers

In [16]:
```
from tensorflow.keras.layers import Dropout                    # Import Dro

# Get information about the class
help(Dropout)
```

```
Help on class Dropout in module keras.src.layers.regularization.dropout:

class Dropout(keras.src.layers.layer.Layer)
 |  Dropout(rate, noise_shape=None, seed=None, **kwargs)
 |
 |  Applies dropout to the input.
 |
 |  The `Dropout` layer randomly sets input units to 0 with a frequency of
 |  `rate` at each step during training time, which helps prevent overfittin
g.
 |  Inputs not set to 0 are scaled up by `1 / (1 - rate)` such that the sum
over
 |  all inputs is unchanged.
 |
 |  Note that the `Dropout` layer only applies when `training` is set to `Tr
ue`
 |  in `call()`, such that no values are dropped during inference.
 |  When using `model.fit`, `training` will be appropriately set to `True`
 |  automatically. In other contexts, you can set the argument explicitly
 |  to `True` when calling the layer.
 |
 |  (This is in contrast to setting `trainable=False` for a `Dropout` layer.
 |  `trainable` does not affect the layer's behavior, as `Dropout` does
 |  not have any variables/weights that can be frozen during training.)
 |
 |  Args:
 |      rate: Float between 0 and 1. Fraction of the input units to drop.
 |      noise_shape: 1D integer tensor representing the shape of the
 |          binary dropout mask that will be multiplied with the input.
 |          For instance, if your inputs have shape
 |          `(batch_size, timesteps, features)` and
 |          you want the dropout mask to be the same for all timesteps,
 |          you can use `noise_shape=(batch_size, 1, features)`.
 |      seed: A Python integer to use as random seed.
 |
 |  Call arguments:
 |      inputs: Input tensor (of any rank).
 |      training: Python boolean indicating whether the layer should behave
in
 |          training mode (adding dropout) or in inference mode (doing nothi
ng).
 |
 |  Method resolution order:
 |      Dropout
 |      keras.src.layers.layer.Layer
 |      keras.src.backend.tensorflow.layer.TFLayer
 |      keras.src.backend.tensorflow.trackable.KerasAutoTrackable
 |      tensorflow.python.trackable.autotrackable.AutoTrackable
 |      tensorflow.python.trackable.base.Trackable
 |      keras.src.ops.operation.Operation
 |      keras.src.saving.keras_saveable.KerasSaveable
 |      builtins.object
 |
 |  Methods defined here:
 |
 |  __init__(self, rate, noise_shape=None, seed=None, **kwargs)
```

```
|      Initialize self.  See help(type(self)) for accurate signature.
|
|  call(self, inputs, training=False)
|
|  compute_output_shape(self, input_shape)
|
|  get_config(self)
|      Returns the config of the object.
|
|      An object config is a Python dictionary (serializable)
|      containing the information needed to re-instantiate it.
|
|  ----------------------------------------------------------------------
|  Methods inherited from keras.src.layers.layer.Layer:
|
|  __call__(self, *args, **kwargs)
|      Call self as a function.
|
|  __delattr__(self, name)
|      Implement delattr(self, name).
|
|  __repr__(self)
|      Return repr(self).
|
|  __setattr__(self, name, value)
|      Support self.foo = trackable syntax.
|
|  __str__(self)
|      Return str(self).
|
|  add_loss(self, loss)
|      Can be called inside of the `call()` method to add a scalar loss.
|
|      Example:
|
|      ```python
|      class MyLayer(Layer):
|          ...
|          def call(self, x):
|              self.add_loss(ops.sum(x))
|              return x
|      ```
|
|  add_metric(self, *args, **kwargs)
|
|  add_variable(
|      self,
|      shape,
|      initializer,
|      dtype=None,
|      trainable=True,
|      autocast=True,
|      regularizer=None,
|      constraint=None,
|      layout=None,
|      name=None
```

```
 |  )
 |      Add a weight variable to the layer.
 |
 |      Alias of `add_weight()`.
 |
 |  add_weight(
 |      self,
 |      *args,
 |      shape=None,
 |      initializer=None,
 |      dtype=None,
 |      trainable=True,
 |      autocast=True,
 |      regularizer=None,
 |      constraint=None,
 |      aggregation='none',
 |      overwrite_with_gradient=False,
 |      layout=None,
 |      name=None
 |  )
 |      Add a weight variable to the layer.
 |
 |      Args:
 |          shape: Shape tuple for the variable. Must be fully-defined
 |              (no `None` entries). Defaults to `()` (scalar) if unspecifie
d.
 |          initializer: Initializer object to use to populate the initial
 |              variable value, or string name of a built-in initializer
 |              (e.g. `"random_normal"`). If unspecified, defaults to
 |              `"glorot_uniform"` for floating-point variables and to `"zer
os"`
 |              for all other types (e.g. int, bool).
 |          dtype: Dtype of the variable to create, e.g. `"float32"`. If
 |              unspecified, defaults to the layer's variable dtype
 |              (which itself defaults to `"float32"` if unspecified).
 |          trainable: Boolean, whether the variable should be trainable via
 |              backprop or whether its updates are managed manually. Defaul
ts
 |              to `True`.
 |          autocast: Boolean, whether to autocast layers variables when
 |              accessing them. Defaults to `True`.
 |          regularizer: Regularizer object to call to apply penalty on the
 |              weight. These penalties are summed into the loss function
 |              during optimization. Defaults to `None`.
 |          constraint: Contrainst object to call on the variable after any
 |              optimizer update, or string name of a built-in constraint.
 |              Defaults to `None`.
 |          aggregation: Optional string, one of `None`, `"none"`, `"mean"`,
 |              `"sum"` or `"only_first_replica"`. Annotates the variable wi
th
 |              the type of multi-replica aggregation to be used for this
 |              variable when writing custom data parallel training loops.
 |              Defaults to `"none"`.
 |          overwrite_with_gradient: Boolean, whether to overwrite the varia
ble
 |              with the computed gradient. This is useful for float8 traini
```

```
ng.
 |              Defaults to `False`.
 |          layout: Optional. Sharding layout for the variable, used instead
of
 |              the one the active Distribution would assign. Either a
 |              `keras.distribution.TensorLayout` with its `device_mesh` se
t, or
 |              a backend-specific layout.
 |          name: String name of the variable. Useful for debugging purpose
s.
 |
 |  build(self, input_shape)
 |
 |  build_from_config(self, config)
 |      Builds the layer's states with the supplied config dict.
 |
 |      By default, this method calls the `build(config["input_shape"])` met
hod,
 |      which creates weights based on the layer's input shape in the suppli
ed
 |      config. If your config contains other information needed to load the
 |      layer's state, you should override this method.
 |
 |      Args:
 |          config: Dict containing the input shape associated with this lay
er.
 |
 |  compute_mask(self, inputs, previous_mask)
 |
 |  compute_output_spec(self, *args, **kwargs)
 |
 |  count_params(self)
 |      Count the total number of scalars composing the weights.
 |
 |      Returns:
 |          An integer count.
 |
 |  get_build_config(self)
 |      Returns a dictionary with the layer's input shape.
 |
 |      This method returns a config dict that can be used by
 |      `build_from_config(config)` to create all states (e.g. Variables and
 |      Lookup tables) needed by the layer.
 |
 |      By default, the config only contains the input shape that the layer
 |      was built with. If you're writing a custom layer that creates state
in
 |      an unusual way, you should override this method to make sure this st
ate
 |      is already created when Keras attempts to load its value upon model
 |      loading.
 |
 |      Returns:
 |          A dict containing the input shape associated with the layer.
 |
 |  get_weights(self)
```

```
 |      Return the values of `layer.weights` as a list of NumPy arrays.
 |
 |  load_own_variables(self, store)
 |      Loads the state of the layer.
 |
 |      You can override this method to take full control of how the state o
f
 |      the layer is loaded upon calling `keras.models.load_model()`.
 |
 |      Args:
 |          store: Dict from which the state of the model will be loaded.
 |
 |  quantize(self, mode=None, type_check=True, config=None)
 |      Quantizes this layer in place.
 |
 |      Resolves `mode`/`config` into a `QuantizationConfig`, validates that
 |      the layer supports the mode *before* mutating any state, then lets
 |      the mode's strategy compute and swap the variables and updates the
 |      dtype policy.
 |
 |      A layer supports quantization by defining
 |      `_quantization_geometry()`; only instances of the exact class that
 |      defines it can be quantized (pass `type_check=False` to quantize an
 |      instance of a subclass that inherits the geometry).
 |
 |      Args:
 |          mode: The quantization mode, e.g. `"int8"`. Optional if `config`
 |              is provided.
 |          type_check: Whether to reject subclasses of the class that owns
 |              the quantization support. Defaults to `True`.
 |          config: Optional `QuantizationConfig` carrying the mode and its
 |              parameters.
 |
 |  quantized_build(self, input_shape, mode, config=None)
 |
 |  quantized_call(self, *args, **kwargs)
 |
 |  rematerialized_call(self, layer_call, *args, **kwargs)
 |      Enable rematerialization dynamically for layer's call method.
 |
 |      Args:
 |          layer_call: The original `call` method of a layer.
 |
 |      Returns:
 |          Rematerialized layer's `call` method.
 |
 |  save_own_variables(self, store)
 |      Saves the state of the layer.
 |
 |      You can override this method to take full control of how the state o
f
 |      the layer is saved upon calling `model.save()`.
 |
 |      Args:
 |          store: Dict where the state of the model will be saved.
 |
```

```
 |  set_weights(self, weights)
 |      Sets the values of `layer.weights` from a list of NumPy arrays.
 |
 |  stateless_call(
 |      self,
 |      trainable_variables,
 |      non_trainable_variables,
 |      *args,
 |      return_losses=False,
 |      **kwargs
 |  )
 |      Call the layer without any side effects.
 |
 |      Args:
 |          trainable_variables: List of trainable variables of the model.
 |          non_trainable_variables: List of non-trainable variables of the
 |              model.
 |          *args: Positional arguments to be passed to `call()`.
 |          return_losses: If `True`, `stateless_call()` will return the lis
t of
 |              losses created during `call()` as part of its return values.
 |          **kwargs: Keyword arguments to be passed to `call()`.
 |
 |      Returns:
 |          A tuple. By default, returns `(outputs, non_trainable_variables)
`.
 |              If `return_losses = True`, then returns
 |              `(outputs, non_trainable_variables, losses)`.
 |
 |      Note: `non_trainable_variables` include not only non-trainable weigh
ts
 |      such as `BatchNormalization` statistics, but also RNG seed state
 |      (if there are any random operations part of the layer, such as dropo
ut),
 |      and `Metric` state (if there are any metrics attached to the layer).
 |      These are all elements of state of the layer.
 |
 |      Example:
 |
 |      ```python
 |      model = ...
 |      data = ...
 |      trainable_variables = model.trainable_variables
 |      non_trainable_variables = model.non_trainable_variables
 |      # Call the model with zero side effects
 |      outputs, non_trainable_variables = model.stateless_call(
 |          trainable_variables,
 |          non_trainable_variables,
 |          data,
 |      )
 |      # Attach the updated state to the model
 |      # (until you do this, the model is still in its pre-call state).
 |      for ref_var, value in zip(
 |          model.non_trainable_variables, non_trainable_variables
 |      ):
 |          ref_var.assign(value)
```

```
    |
    |  symbolic_call(self, *args, **kwargs)
    |
    |  ----------------------------------------------------------------------
    |  Static methods inherited from keras.src.layers.layer.Layer:
    |
    |  __new__(cls, *args, **kwargs)
    |      Create and return a new object.  See help(type) for accurate signatu
re.
    |
    |  ----------------------------------------------------------------------
    |  Readonly properties inherited from keras.src.layers.layer.Layer:
    |
    |  compute_dtype
    |      The dtype of the computations performed by the layer.
    |
    |  dtype
    |      Alias of `layer.variable_dtype`.
    |
    |  input_dtype
    |      The dtype layer inputs should be converted to.
    |
    |  losses
    |      List of scalar losses from `add_loss`, regularizers and sublayers.
    |
    |  metrics
    |      List of all metrics.
    |
    |  metrics_variables
    |      List of all metric variables.
    |
    |  non_trainable_variables
    |      List of all non-trainable layer state.
    |
    |      This extends `layer.non_trainable_weights` to include all state used
by
    |      the layer including state for metrics and `SeedGenerator`s.
    |
    |  non_trainable_weights
    |      List of all non-trainable weight variables of the layer.
    |
    |      These are the weights that should not be updated by the optimizer du
ring
    |      training. Unlike, `layer.non_trainable_variables` this excludes metr
ic
    |      state and random seeds.
    |
    |  path
    |      The path of the layer.
    |
    |      If the layer has not been built yet, it will be `None`.
    |
    |  quantization_mode
    |      The quantization mode of this layer, `None` if not quantized.
    |
```

```
 |  trainable_variables
 |      List of all trainable layer state.
 |
 |      This is equivalent to `layer.trainable_weights`.
 |
 |  trainable_weights
 |      List of all trainable weight variables of the layer.
 |
 |      These are the weights that get updated by the optimizer during train
ing.
 |
 |  variable_dtype
 |      The dtype of the state (weights) of the layer.
 |
 |  variable_serialization_spec
 |      Maps each supported quantization mode to its variable names.
 |
 |      A quantizable layer returns a dict from mode name (`None` for the
 |      float layer) to the ordered names its `save_own_variables` and
 |      `load_own_variables` serialize in that mode. The base implementation
 |      returns `None`: the layer has no quantization support.
 |
 |      Returns:
 |          `None`, or a dict such as
 |
 |          ```python
 |          {
 |              None: ["kernel", "bias"],
 |              "int8": ["kernel", "bias", "kernel_scale"],
 |          }
 |          ```
 |
 |  variables
 |      List of all layer state, including random seeds.
 |
 |      This extends `layer.weights` to include all state used by the layer
 |      including `SeedGenerator`s.
 |
 |      Note that metrics variables are not included here, use
 |      `metrics_variables` to visit all the metric variables.
 |
 |  weights
 |      List of all weight variables of the layer.
 |
 |      Unlike, `layer.variables` this excludes metric state and random seed
s.
 |
 |  ----------------------------------------------------------------------
 |  Data descriptors inherited from keras.src.layers.layer.Layer:
 |
 |  dtype_policy
 |
 |  input_spec
 |
 |  supports_masking
 |      Whether this layer supports computing a mask using `compute_mask`.
```

```
 |
 |  trainable
 |      Settable boolean, whether this layer should be trainable or not.
 |
 |  ----------------------------------------------------------------------
 |  Data and other attributes inherited from keras.src.layers.layer.Layer:
 |
 |  lora_enabled = False
 |
 |  ----------------------------------------------------------------------
 |  Data descriptors inherited from tensorflow.python.trackable.base.Trackab
le:
 |
 |  __dict__
 |      dictionary for instance variables
 |
 |  __weakref__
 |      list of weak references to the object
 |
 |  ----------------------------------------------------------------------
 |  Class methods inherited from keras.src.ops.operation.Operation:
 |
 |  from_config(config)
 |      Creates an operation from its config.
 |
 |      This method is the reverse of `get_config`, capable of instantiating
the
 |      same operation from the config dictionary.
 |
 |      Note: If you override this method, you might receive a serialized dt
ype
 |      config, which is a `dict`. You can deserialize it as follows:
 |
 |      ```python
 |      if "dtype" in config and isinstance(config["dtype"], dict):
 |          policy = dtype_policies.deserialize(config["dtype"])
 |      ```
 |
 |      Args:
 |          config: A Python dictionary, typically the output of `get_config
`.
 |
 |      Returns:
 |          An operation instance.
 |
 |  ----------------------------------------------------------------------
 |  Readonly properties inherited from keras.src.ops.operation.Operation:
 |
 |  input
 |      Retrieves the input tensor(s) of a symbolic operation.
 |
 |      Only returns the tensor(s) corresponding to the *first time*
 |      the operation was called.
 |
 |      Returns:
 |          Input tensor or list of input tensors.
```

```
 |
 |  output
 |      Retrieves the output tensor(s) of a layer.
 |
 |      Only returns the tensor(s) corresponding to the *first time*
 |      the operation was called.
 |
 |      Returns:
 |          Output tensor or list of output tensors.
 |
 |  ----------------------------------------------------------------------
 |  Methods inherited from keras.src.saving.keras_saveable.KerasSaveable:
 |
 |  __reduce__(self)
 |      __reduce__ is used to customize the behavior of `pickle.pickle()`.
 |
 |      The method returns a tuple of two elements: a function, and a list o
f
 |      arguments to pass to that function.  In this case we just leverage t
he
 |      keras saving library.
```

In [ ]:
```
# we delete some neurons randomly at each iteration

# Train again the new model
model = Sequential()

model.add(Dense(units=30,activation='relu'))
model.add(Dropout(0.5))                                    # Add Dropout opt

model.add(Dense(units=15,activation='relu'))
model.add(Dropout(0.5))                                    # Add Dropout opt

model.add(Dense(units=1,activation='sigmoid'))

model.compile(loss='binary_crossentropy', optimizer='adam')
```

In [18]:
```
model.fit(x=X_train, y=y_train, epochs=600, validation_data=(X_test, y_test)
```

```
Epoch 1/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 38ms/step - loss: 0.7705 - val_loss: 0.5038
Epoch 2/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.6150 - val_loss: 0.4193
Epoch 3/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.5246 - val_loss: 0.3640
Epoch 4/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.4419 - val_loss: 0.3209
Epoch 5/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.4738 - val_loss: 0.2850
Epoch 6/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.3748 - val_loss: 0.2600
Epoch 7/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.3184 - val_loss: 0.2369
Epoch 8/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.3165 - val_loss: 0.2162
Epoch 9/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.3215 - val_loss: 0.2001
Epoch 10/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 0.3011 - val_loss: 0.1862
Epoch 11/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.2894 - val_loss: 0.1722
Epoch 12/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.2561 - val_loss: 0.1610
Epoch 13/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.2402 - val_loss: 0.1521
Epoch 14/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.2158 - val_loss: 0.1440
Epoch 15/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.2332 - val_loss: 0.1379
Epoch 16/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.2161 - val_loss: 0.1308
Epoch 17/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1933 - val_loss: 0.1242
Epoch 18/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.2164 - val_loss: 0.1187
Epoch 19/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.2094 - val_loss: 0.1147
Epoch 20/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1831 - val_loss: 0.1109
Epoch 21/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1983 - val_loss: 0.1075
Epoch 22/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.2222 - val_loss: 0.1052
Epoch 23/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1587 - val_loss: 0.1026
Epoch 24/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - loss: 0.1343 - val_loss: 0.0997
Epoch 25/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1486 - val_loss: 0.0967
Epoch 26/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1516 - val_loss: 0.0942
Epoch 27/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1650 - val_loss: 0.0920
Epoch 28/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1583 - val_loss: 0.0904
```

```
Epoch 29/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1803 - val_loss: 0.0891
Epoch 30/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1481 - val_loss: 0.0883
Epoch 31/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1509 - val_loss: 0.0877
Epoch 32/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1638 - val_loss: 0.0875
Epoch 33/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1353 - val_loss: 0.0869
Epoch 34/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1587 - val_loss: 0.0868
Epoch 35/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1413 - val_loss: 0.0858
Epoch 36/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1375 - val_loss: 0.0855
Epoch 37/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1311 - val_loss: 0.0851
Epoch 38/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1092 - val_loss: 0.0840
Epoch 39/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1181 - val_loss: 0.0837
Epoch 40/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1424 - val_loss: 0.0836
Epoch 41/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1403 - val_loss: 0.0838
Epoch 42/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0961 - val_loss: 0.0833
Epoch 43/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1279 - val_loss: 0.0835
Epoch 44/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1106 - val_loss: 0.0828
Epoch 45/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0974 - val_loss: 0.0819
Epoch 46/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1029 - val_loss: 0.0812
Epoch 47/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1313 - val_loss: 0.0806
Epoch 48/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1193 - val_loss: 0.0808
Epoch 49/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0893 - val_loss: 0.0806
Epoch 50/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.1049 - val_loss: 0.0804
Epoch 51/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1189 - val_loss: 0.0805
Epoch 52/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0892 - val_loss: 0.0803
Epoch 53/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0858 - val_loss: 0.0806
Epoch 54/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1000 - val_loss: 0.0803
Epoch 55/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1142 - val_loss: 0.0810
Epoch 56/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0966 - val_loss: 0.0817
```

```
Epoch 57/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1130 - val_loss: 0.0822
Epoch 58/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0990 - val_loss: 0.0835
Epoch 59/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1063 - val_loss: 0.0848
Epoch 60/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.1147 - val_loss: 0.0859
Epoch 61/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0915 - val_loss: 0.0864
Epoch 62/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0874 - val_loss: 0.0863
Epoch 63/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.1134 - val_loss: 0.0872
Epoch 64/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0770 - val_loss: 0.0878
Epoch 65/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0878 - val_loss: 0.0890
Epoch 66/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0929 - val_loss: 0.0893
Epoch 67/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0865 - val_loss: 0.0887
Epoch 68/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0810 - val_loss: 0.0887
Epoch 69/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0956 - val_loss: 0.0897
Epoch 70/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.0895 - val_loss: 0.0902
Epoch 71/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0719 - val_loss: 0.0901
Epoch 72/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 0.0785 - val_loss: 0.0897
Epoch 73/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 0.1176 - val_loss: 0.0884
Epoch 74/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0903 - val_loss: 0.0866
Epoch 75/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.1001 - val_loss: 0.0872
Epoch 76/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0853 - val_loss: 0.0886
Epoch 77/600
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.0839 - val_loss: 0.0900
Epoch 77: early stopping
```

Out[18]: <keras.src.callbacks.history.History at 0x7fc520bc42d0>

In [19]:
```
# Get information about model
model.summary()

# Save Training and Validation Loss in a Pandas DataFramemodel_loss.plot()
model_loss = pd.DataFrame(model.history.history)

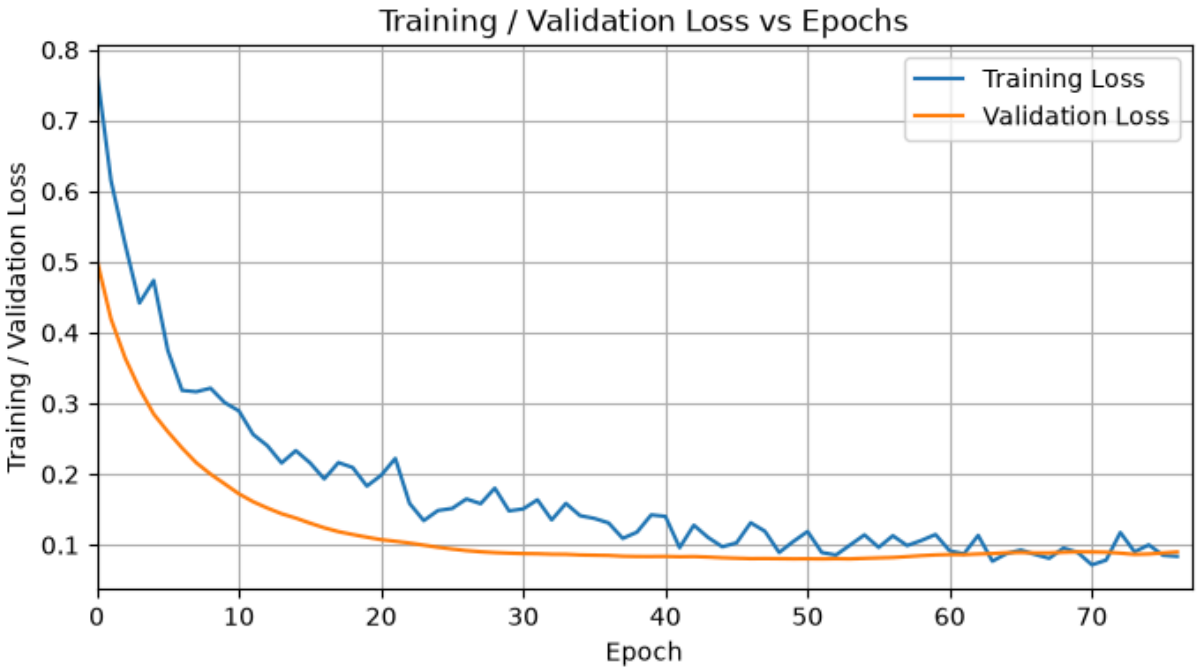
# Plot Training / Validation Loss vs Epochs
plt.figure(figsize = (8, 4))
plt.plot(range(len(model_loss)), model_loss['loss'], label = 'Training Loss'
plt.plot(range(len(model_loss)), model_loss['val_loss'], label = 'Validatior
```

```
plt.xlim([0, len(model_loss)])
plt.xlabel('Epoch')
plt.ylabel('Training / Validation Loss')
plt.title('Training / Validation Loss vs Epochs')
plt.legend()
plt.grid()
```

Model: "sequential_2"

Layer (type)	Output Shape	Par
dense_6 (Dense)	(32, 30)	
dropout (Dropout)	(32, 30)	
dense_7 (Dense)	(32, 15)	
dropout_1 (Dropout)	(32, 15)	
dense_8 (Dense)	(32, 1)	

Total params: 4,235 (16.55 KB)
Trainable params: 1,411 (5.51 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 2,824 (11.04 KB)



Model Evaluation

In [20]:
```
from sklearn.metrics import classification_report, confusion_matrix

# Get predictions and set them to either 1 or 0
predictions = model.predict(X_test)
```

```python
predictions = np.where(predictions > 0.5, 1, 0)

# Print results
print(classification_report(y_test, predictions))
print('Confusion matrix')
print(confusion_matrix(y_test, predictions))
```

```
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step
              precision    recall  f1-score   support

           0       0.96      0.96      0.96        55
           1       0.98      0.98      0.98        88

    accuracy                           0.97       143
   macro avg       0.97      0.97      0.97       143
weighted avg       0.97      0.97      0.97       143

Confusion matrix
[[53  2]
 [ 2 86]]
```